

Enhancing air quality forecasting through missing data imputation: a stacking-based approach applied to urban monitoring data

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ABSTRACT

In air pollution forecasting, handling missing data is a significant challenge and a crucial task. In this study, we aim to forecast air pollutants in Eskişehir, Türkiye, presenting a new stacking-based imputation method (S-UImpute) to address the missing data problem. To this end, we apply three widely used forecasting methods: Autoregressive Integrated Moving Average (ARIMA), Support Vector Regression (SVR), and Long Short-Term Memory (LSTM), and eight commonly used missing data imputation methods (ZeroFill, MeanFill, MedianFill, ForwardFill, BackwardFill, Interpolation, Kalman filtering, and SVR-based imputation) for benchmarking on univariate time series data. We collect comprehensive datasets, which have various proportions of missing data, from five air quality monitoring stations in Eskişehir. To evaluate the performance of the proposed method at high missing data rates, we generated artificial missing data on top of the existing missing data at rates of 10%, 15%, 20%, and 25%. The S-UImpute + LSTM model mostly outperformed the other imputation method and forecasting model combinations in terms of modified symmetric Mean Absolute Percentage Error (msMAPE). Moreover, S-UImpute provided more consistent and generalized performance compared with other imputation methods. As LSTM produced the most accurate forecasting, it is used to statistically compare imputation methods and determine the best-performing one. According to the Friedman statistical test, there is a statistically significant difference between the missing data imputation methods. Following this, the Wilcoxon signed-rank test applied afterwards indicates that the best method is S-UImpute.

Subjects Artificial Intelligence, Computer Education, Data Mining and Machine Learning, Data Science

Keywords Missing data, Air quality, Stacking ensemble, Univariate time series

INTRODUCTION

A silent killer, air pollution is an invisible but constant threat that affects millions of people every day without their awareness. According to the [World Health Organization \(2025\)](#) (WHO), 99% of the world's population breathes polluted air that exceeds WHO guideline limits. Middle-income countries are the most affected by this situation. Türkiye, as an upper-middle-income country, is also significantly affected by air pollution, especially in urban and industrialized areas.

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In Eskişehir, traffic, industrial activities, and domestic heating significantly contribute to air pollution, making effective pollution control and policy development necessary (Balçılar, 2024). Moreover, previous studies have demonstrated a connection between air quality and respiratory diseases (Tağıl & Menteşe, 2012). Statistical data indicate that 12–14% of deaths in Eskişehir between 2020–2023 were due to respiratory diseases (Turkish Statistical Institute, 2023; Kangas et al., 2023). In light of these, effective air quality prediction models are critical for pollution control and policy development for Eskişehir.

Since air quality datasets collected over extended periods often suffer from missing values, leading to sparsity, missing data handling is crucial in the preprocessing step (Zhang et al., 2024). The use of low-cost Internet of Things (IoT) sensors may also cause measurement failures or inaccuracies, resulting in missing values and reducing the reliability of forecasting models (Essamlali, Nhaila & El Khaili, 2024).

To handle the missing data, imputation or deletion can be applied as a preprocessing step. However, a review by Yu et al. (2024) found that only 10% of air-quality prediction studies address missing data processing, showing that it is often neglected in the literature. Imputation in univariate time series is particularly challenging. Most imputation algorithms use feature correlations to predict missing data (Moritz et al., 2015). However, there are no additional features and correlations in univariate time series. On the other hand, since air quality data is a time series, deleting any missing information might distort the data structure and make time series analysis more difficult (Zhou et al., 2021).

This study aims to forecast air pollutant levels in Eskişehir, Türkiye. For this purpose, we used air pollutant data collected from five air quality stations located within the borders of Eskişehir. The datasets contained varying rates of missing data. In this study, we propose a new stacking-based imputation method to preserve the structure of air pollutant time series and eliminate the missing data problem. Unlike conventional single method approaches, the novel imputation method developed is a combination of both existing imputation methods and machine learning models. Thus, we developed a novel imputation method that generalizes the prediction performances of existing imputation methods. The main contributions of this study are as follows:

- A novel stacking-based imputation method (S-UImpute) is proposed, combining existing imputation methods and machine learning models in a stacking ensemble framework.
- S-UImpute improves robustness by capturing temporal patterns and generalizing over individual imputation methods.
- S-UImpute is validated on real-world air pollutant data from five air quality monitoring stations located in Eskişehir, Türkiye, under different missingness rates.
- The effect of S-UImpute on forecasting accuracy is evaluated using three different models: Autoregressive Integrated Moving Average (ARIMA), Support Vector Regression (SVR), and Long Short-Term Memory (LSTM). Results show

that S-UImpute notably improves forecasting performance, mainly when used with LSTM.

From a theoretical perspective, this study contributes to the literature by proposing a stacking-based imputation method, S-UImpute, for handling missing data in univariate air-quality time series. Unlike traditional approaches that are based on a single imputation method, S-UImpute integrates multiple imputation methods within a stacking ensemble structure to improve robustness and stability across different missingness rates. In addition, this study provides a systematic evaluation of how different imputation methods affect downstream forecasting performance.

From a practical perspective, the proposed approach can be useful for real-world air-quality monitoring systems where missing observations frequently occur due to sensor failures or measurement problems. Empirical analysis of real-world univariate time-series datasets from five air-quality monitoring stations shows that S-UImpute improves the forecasting performance of deep learning models, particularly LSTM. Therefore, the proposed approach can serve as an effective preprocessing step to improve the reliability of air-quality forecasting and support environmental decision-making.

The rest of the study is organized as follows: The literature about missing data imputation in air quality studies is summarized in ‘Literature Review’. In ‘Materials and Methods’, the considered datasets are explained in detail, along with the methods for imputing missing data in the time series used in the study, including machine learning models and ensemble models, as well as the proposed S-UImpute. ‘Experimental Study’ presents the experimental research, test results, and commentary on the findings. ‘Discussion’ discusses the findings, describes the strengths and limitations of the proposed approach. The study’s importance, results, and future works are mentioned in ‘Conclusions’.

LITERATURE REVIEW

Air pollution is a significant global problem due to its harmful effects on public health and the environment. Consequently, accurate air quality forecasting has become an important area of research to support early warning and decision-making ([Wang et al., 2024](#)). Many studies have used various methods to forecast air quality, including statistical models (e.g., ARIMA) ([Mani, Viswanadhapalli & Stonier, 2022](#)), machine-learning algorithms (e.g., Random Forest, Gradient Boosting) ([Ravindiran et al., 2023](#)), and deep-learning models such as Convolutional Neural Network (CNN) and LSTM ([Mao et al., 2021](#)). In addition, several recent transformer-based models, such as AirFormer ([Liang et al., 2023](#)), DSFormer ([Yu et al., 2023](#)), and MGSFformer ([Yu et al., 2025](#)), have achieved strong performance in multivariate air-quality forecasting by capturing long-range temporal and spatial dependencies. Preprocessing plays a critical role in air-quality forecasting, as improper handling of missing values, scaling, or feature selection can decrease forecasting performance ([Narkhede et al., 2025](#)). Among these preprocessing steps, handling missing data is especially important because air-quality datasets often contain missing values due

to sensor issues, which can affect forecasting accuracy (Junninen et al., 2004). The remainder of this section focuses on missing data imputation in air quality forecasting.

Studies addressing missing data in air quality forecasting typically involve either univariate or multivariate time series. However, most studies focus on multivariate datasets. Hua et al. (2024) considered air quality prediction and missing data imputation techniques for multivariate time series data. Mean, median, k-Nearest Neighbors Imputation (kNNI), Multiple Imputation by Chained Equations (MICE), Self-attention-based imputation for time series (SAITS), Bidirectional Recurrent Imputation for Time Series (BRITS), Multi-directional Recurrent Neural Network (MRNN), and Transformer imputation methods were applied to five real datasets, with artificially introduced missing at random patterns. Imputation accuracy was used to evaluate the performance of the imputation methods. SAITS gave the lowest imputation error. Alsaber, Pan & Al-Hurban (2021) utilized Missforest, based on Random Forest, to overcome the missing data problem, which involves different missing data mechanisms, and applied their proposed approach for Kuwait air quality measurements, sulfur dioxide (SO_2), nitrogen dioxide (NO_2), carbon monoxide (CO), particulate matter with an aerodynamic diameter of less than 10 micrometers (PM_{10}), and ozone O_3 . Lee et al. (2025) investigated the imputation accuracy of different methods for handling artificially removed particulate matter with an aerodynamic diameter of less than 2.5 micrometers ($\text{PM}_{2.5}$) values using a multivariate dataset that included air pollutants in Korea. Their findings showed that simple imputation methods were more accurate for the short term, while advanced imputation models performed better for the long term. However, in real-world settings, the actual values of missing data are unknown, so imputation accuracy cannot be directly assessed. Therefore, the predictive performance of downstream models becomes a more meaningful metric (Jäger, Allhorn & Bießmann, 2021).

The presence of missing data in time series not only affects data quality but also reduces the predictive performance of models by introducing errors and bias (Tawakuli et al., 2025; Eren, Aksangür & Erden, 2023). Du, Côté & Liu (2023) proposed the self-attention-based SAITS model and showed that it achieves high imputation accuracy for multivariate time-series data and improves classification performance. Umar & Gray (2023) compared single and multiple imputation methods for both univariate and multivariate hydrological time-series data, showing that the choice of imputation technique strongly affects imputation accuracy. Similarly, Niako et al. (2024) showed that different imputation methods yield significant differences in ARIMA and LSTM forecasting performance for univariate blood pressure time-series data. However, few studies have systematically investigated the effects of missing data imputation on air quality prediction models. Alkabbani et al. (2022) used artificial neural networks (ANN) and a multivariate air quality dataset to forecast air quality and compared the effects of MissForest and linear interpolation imputation methods on forecasting performance. MissForest achieved higher accuracy in forecasting.

It has been shown that univariate models can outperform multivariate models in air quality forecasting despite the widespread use of multivariate models. Sethi & Mittal (2020) demonstrated that the univariate model generated more accurate predictions for air

quality than the multivariate model, despite the latter's ability to capture the interdependencies between pollutants and meteorological variables. Similarly, [Bakar et al. \(2024\)](#) revealed that a simpler univariate CNN model yields more accurate forecasts than the multivariate CNN model. Moreover, many monitoring stations collect observations for only a single variable, making univariate models a more practical option in such cases. [Hadeed et al. \(2020\)](#) compared various imputation methods for short-term $PM_{2.5}$ measurements in terms of imputation accuracy but did not investigate their impact on forecasting performance. [Cicek & Aydin \(2025\)](#) examined the effect of commonly used imputation methods on LSTM-based air-quality forecasting. Their findings supported that the interpolation imputation method yielded the best prediction performance for LSTM. However, it has been observed that methods for imputing missing data in univariate air quality forecasting have not been explored sufficiently, highlighting a gap in the existing literature. To address this gap, this study applies various missing data imputation methods for univariate air pollutant forecasting, proposes a novel stacking-based imputation approach, and evaluates its impact on forecasting performance.

MATERIALS AND METHODS

This section introduces the datasets used in the study and then explains the methods for imputing missing data in time series and the forecasting methods employed. We also give a brief explanation of the missing data mechanisms considered in this section. In univariate time series, under the Missing Completely at Random (MCAR) mechanism, the missing data occur completely randomly. In contrast, in the Missing at Random (MAR) mechanism, the missing data depend on observed factors, typically time. In the Missing Not at Random (MNAR) mechanism, the missing data occur depending on their own value. Finally, it also introduces the proposed stacking-based ensemble imputation method, S-UImpute.

Dataset

This study utilized hourly air pollution data collected from five monitoring stations in Eskişehir between 2024 and 2025. The locations of these five stations are shown on the map in [Fig. 1](#). As illustrated in the map, the Metin Sönmez station is located near the organized industrial zone, whereas the other four stations are distributed across residential areas in the city center. The datasets were obtained from the Turkish Ministry of Environment and Urbanization (https://sim.csb.gov.tr/STN/STN_Report/StationDataDownloadNew). The characteristics of the dataset are given in [Table 1](#). Because the datasets are univariate, their upper and lower limits are shown by the [Min, Max] values in the table. Moreover, [Figs. 2](#) and [3](#) visualize the datasets and the missingness map, respectively. At all stations except Metin Sönmez, located near the industrial zone, a noticeable decrease in air pollutant concentrations is observed during the summer. This seasonal change can be attributed to domestic heating in the residential areas where the other stations are located. According to the missingness map in [Fig. 3](#), we observed that there is almost no missing data at the Cumhuriyet Bulvarı station, whereas significant missingness is present across different date

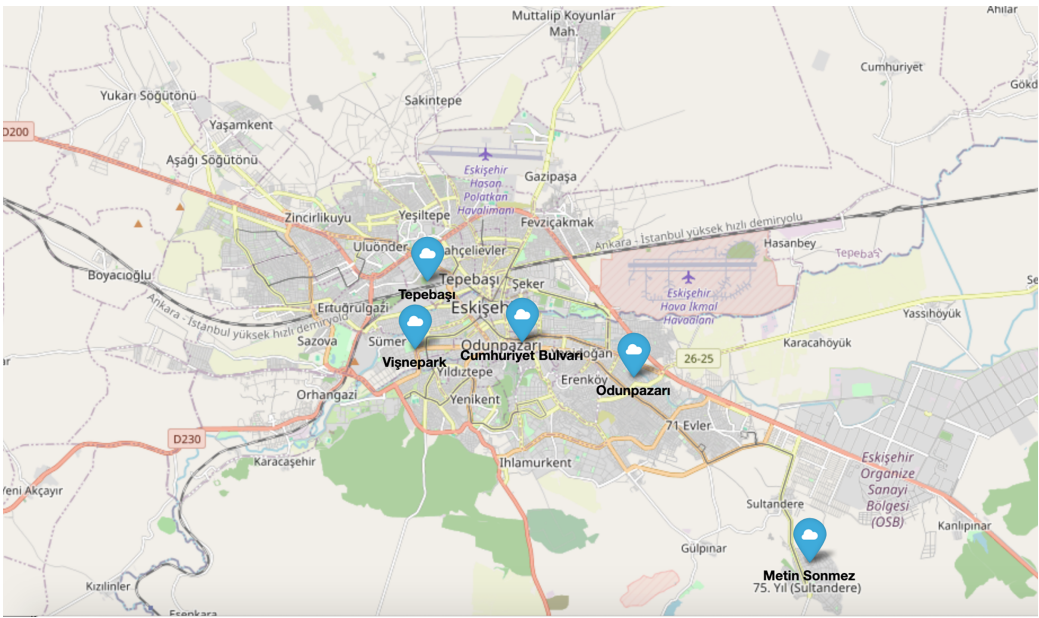


Figure 1 Location of air quality monitoring stations in Eskişehir.
Full-size DOI: 10.7717/peerj-cs.3904/fig-1

Table 1 Dataset characteristics.

Station	Pollutant (µg/m3)	Missing rate (%)	Average	Standard deviation	[Min, Max]
Vişne park	PM ₁₀	6.410	38.455	30.992	[1.000, 995.000]
Metin sönmez	PM _{2,5}	4.432	7.451	8.399	[0.050, 85.810]
Cumhuriyet bulvarı	PM _{2,5}	0.886	22.644	17.266	[0.720, 161.330]
Tepebaşı	PM ₁₀	4.534	46.356	19.324	[7.11, 208.580]
Odunpazarı	PM _{2,5}	2.795	11.923	12.881	[0.010, 148.850]

ranges at the other stations. Additionally, the missing data do not show any hourly pattern at any of the stations.

Missingness is not ignorable under the MNAR mechanism, while it is ignorable under the MCAR or MAR. Standard imputation methods can be appropriately applied under ignorable mechanisms. Although the missing data mechanism can affect the estimation results, previous studies highlight that the difference between MAR and MCAR becomes less significant in univariate time series due to the way time affects the data (Moritz et al., 2015). Accordingly, to evaluate the performance of the proposed method under varying missing data rates, we introduced artificial missingness into the original dataset of each station at rates of 10%, 15%, 20%, and 25% based on the MCAR mechanism. These missing data rates are chosen to evaluate the performance, robustness, and stability of the proposed method under different levels of missingness by simulating missing data rates that are frequently observed in real-world time series datasets. The air pollution datasets, including both the original datasets and the artificially generated missing data versions, are publicly available at our GitHub repository (Çiçek & Aydın, 2025).

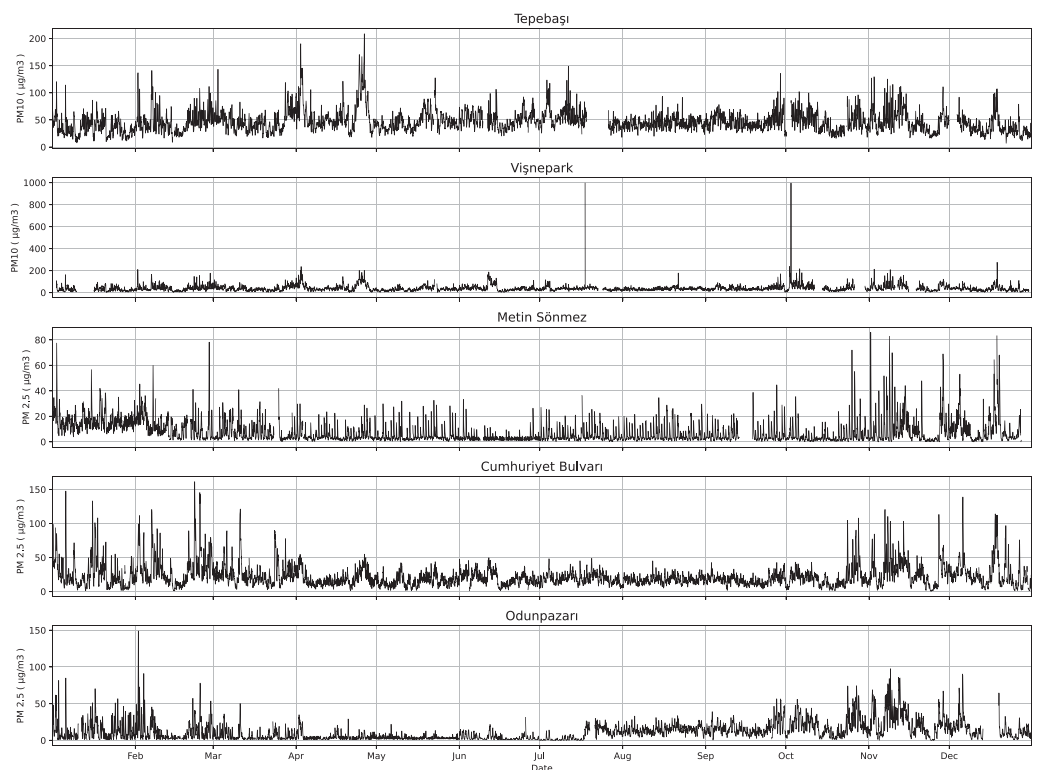


Figure 2 Datasets.

Full-size DOI: [10.7717/peerj-cs.3904/fig-2](https://doi.org/10.7717/peerj-cs.3904/fig-2)

Missing data imputation methods

To evaluate the performance of the proposed S-UImpute method, we compare it with eight commonly used methods for imputing missing data. An overview of these methods is provided below:

- ZeroFill: This method is a replacement strategy in which missing observations are filled with zeros.
- MeanFill: Missing observations are filled with the mean of the known observations.
- MedianFill: This method replaces missing observations using the median value of the existing observations.
- ForwardFill (FFill): Missing observations are replaced sequentially with the last available observation preceding them.
- BackwardFill (BFill): Missing values are replaced with the next available observation in the time series.
- Interpolation: Linear interpolation estimates a missing observation by constructing a straight line between the nearest known values before and after the missing data point. In this study, the term “interpolation” specifically refers to linear interpolation.
- Kalman: Kalman filtering imputes missing observations by estimating the hidden states of the time series based on both past and future values. It uses a mathematical model of the system’s dynamics and measurement uncertainty to produce optimal estimates,

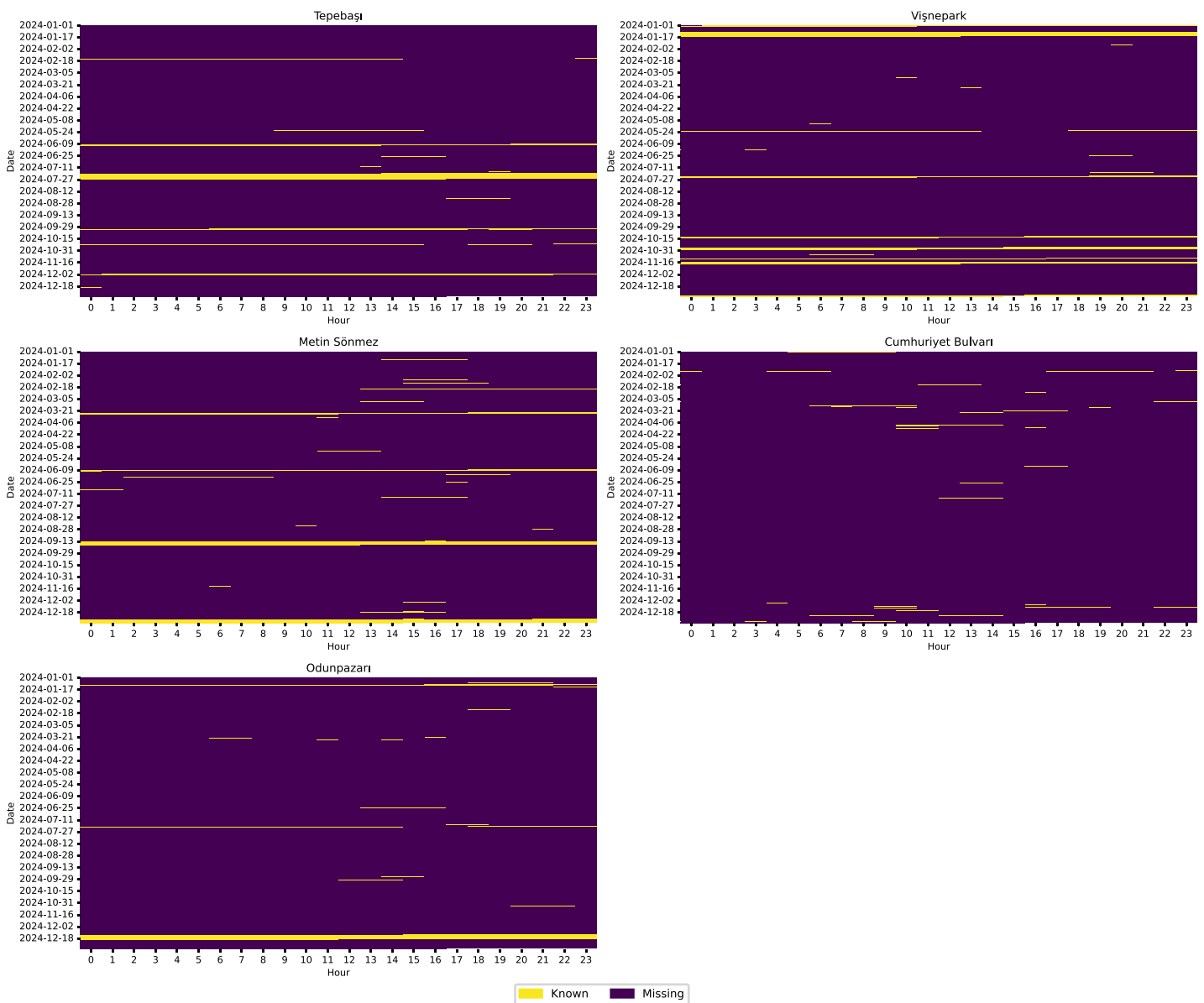


Figure 3 Missingness map of air pollution data in Eskişehir.

Full-size DOI: 10.7717/peerj-cs.3904/fig-3

allowing the reconstruction of missing data in a statistically consistent way (Umar & Gray, 2023).

- Support Vector Regression (SVR): SVR imputes missing observations by estimating them as the dependent variable in a regression model trained on the known observations estimating as independent variables. In univariate time series, there is a single variable; lagged values of that variable are used as independent inputs.

Forecasting methods

Three different forecasting models were applied to observe the effect of the imputation method on forecasting performance. In this context, Autoregressive Integrated Moving

Average (ARIMA) from statistical methods, Support Vector Regression (SVR) from machine learning methods, and Long Short-Term Memory (LSTM) from deep learning methods were selected due to their distinct structures and capabilities. According to a recent survey by [Méndez, Merayo & Núñez \(2023\)](#), these models are also among the most commonly used techniques in air quality forecasting, which further supports their selection in this study.

ARIMA is a widely used statistical method in time-series forecasting. The ARIMA method proposed by [Box & Jenkins \(1970\)](#) consists of three parts. The first part, which is the degree of autocorrelation of the time series expressed in lagged terms, is called the Autoregressive (AR) model. The second part that is expressed with residuals is the Moving Average (MA) model. The expression of Integrated (I) refers to the differencing process applied to a non-stationary model to make it stationary. An ARIMA model contains three parameters: p , d , and q . While p and q are the orders of the AR and MA models, respectively, d refers to the number of times the differencing process is applied. A general ARIMA (p, d, q) model is expressed as follows ([Hyndman & Athanasopoulos, 2021](#)):

$$y'_t = c + \phi_1 y'_{t-1} + \dots + \phi_p y'_{t-p} + \theta_1 \varepsilon_{t-1} + \dots + \theta_q \varepsilon_{t-q} + \varepsilon_t \quad (1)$$

where y'_t denotes differenced series, ε_t are residuals. c is the constant term, the θ and ϕ coefficients refer to the regression coefficients.

Support Vector Regression (SVR) is a variant of Support Vector Machines (SVM), one of the most popular machine learning algorithms. SVM was originally introduced for classification problems ([Cortes & Vapnik, 1995](#)), and was later extended to regression tasks through the SVR framework proposed by [Drucker et al. \(1996\)](#). For the training set $D = \{x_i, y_i\}_{i=1}^n$, x_i denotes the independent variable, where y_i is the dependent variable. SVR trains a regression model using this training set and then utilizes the trained model to predict future target values. The SVR function is given in [Eq. \(2\)](#):

$$f(x) = w^T x + b \quad (2)$$

where w and b are the parameters of the regression model.

The following minimization problem yields to these parameters of the SVR function given in [Eq. \(2\)](#):

$$\min \frac{1}{2} \|w\|^2 + C \sum_{i=1}^n (\xi_i + \xi_i^*). \quad (3)$$

$$y_i - w^T x_i - b \leq \varepsilon + \xi_i \quad \forall i = 1, \dots, n. \quad (4)$$

$$w^T x_i + b - y_i \leq \varepsilon + \xi_i^* \quad \forall i = 1, \dots, n. \quad (5)$$

$$\xi_i \geq 0, \quad \xi_i^* \geq 0 \quad \forall i = 1, \dots, n. \quad (6)$$

In the mathematical model of the SVR, ε denotes the epsilon-insensitive margin within which errors are not penalized. The slack variables ξ_i and ξ_i^* allow for deviations beyond this margin by quantifying the amount of excess error. The parameter C controls the trade-off between model complexity and the penalty imposed on errors that exceed the ε margin.

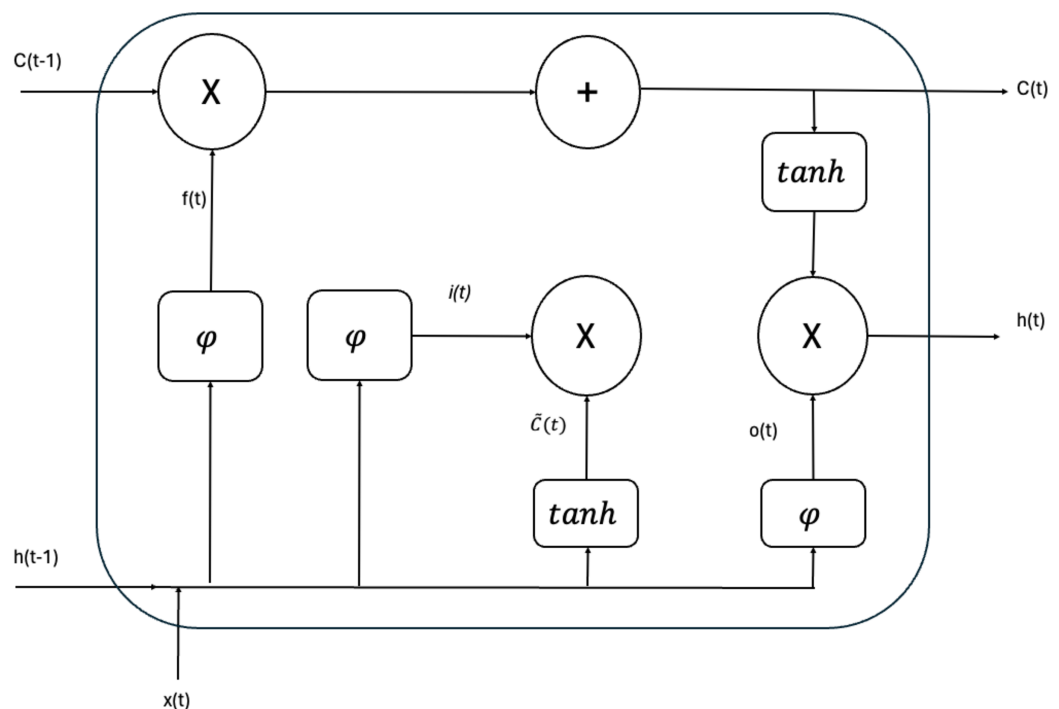


Figure 4 A LSTM cell.

Full-size DOI: 10.7717/peerj-cs.3904/fig-4

LSTM, which was proposed by [Hochreiter & Schmidhuber \(1997\)](#), is an extension of a popular deep learning method, Recurrent Neural Networks. A phenomenon known as the “vanishing gradient problem” occurs during the training of a deep neural network. As they are backpropagated from the output layers to the previous layers, the gradients that are used to update the network weights get small and can be thought of as “vanishing”. To overcome the vanishing gradient problem, LSTM utilizes memory cells, also known as cell states, which store information over extended periods. This makes LSTM work well on sequential data-related issues, such as time series forecasting, language modeling, and machine translation.

Gates and a cell state are the two primary components of an LSTM. The memory cell, also known as the cell state, can add or remove information from the cell state and functions similarly to a conveyor belt ([Cicek, 2024](#)). LSTM cells use “forget,” “input,” and “output” gates to control the flow of information. Because these gates regulate which information is remembered and which is forgotten, the model can recall long-term context. [Figure 4](#) depicts a simple LSTM cell ([Diniz et al., 2024](#)).

In [Fig. 4](#), x_t and h_t denote the input and output of the LSTM cell at time t , respectively; h_{t-1} is the output of the previous time step. c_t and c_{t-1} refer to the cell state at times t and $t - 1$. i_t , o_t , f_t refer to the input gate, output gate, and the forget gate. Lastly, φ is a nonlinear activation function, generally a sigmoid function. During training, the internal states of LSTM cells are updated interactively across time steps, where the hidden and cell states, respectively, denote short-term and long-term dependencies. Gate mechanisms regulate the flow of information by adding or removing content, enabling the network to

retain or discard temporal features as needed. These internal states are reset after each batch ([Diniz et al., 2024](#)).

Stacking ensemble model

Ensemble learning is basically a machine learning paradigm that combines multiple different models to achieve better performance. Ensemble methods enhance the model's performance, enabling it to produce superior predictions compared to a single model, and also strengthen the model's robustness by reducing the dispersion of predictions and performance ([Saihood & Sonuç, 2023](#)).

The most commonly used methods to implement ensemble learning are Bagging, Boosting, and Stacking. Bagging, also known as bootstrap aggregation, is a popular ensemble technique that enhances model performance and stability by combining multiple parallel versions of the same base learner ([Saihood & Sonuç, 2023](#)). While each of the individual base learners is trained with different randomly selected subsets of the training set in the bootstrapping phase, in the aggregation, the predictions are performed using these base learners, and a voting operation is performed to determine the final prediction. To reduce bias and improve prediction accuracy, boosting trains models sequentially. Each new model is trained to fix the errors made by the previously trained models. The final prediction is a weighted combination of all models.

Stacking, which is applied to improve the performance of individual models, uses an ensemble mechanism that uses the output of individual models instead of training data ([Aleryani, Wang & De La Iglesia, 2020](#)). A stacking model typically has two layers. The base models are located in the first layer, known as Level 0 or weak learners, whereas the model in the Level 1 or meta-learner layer is trained using the predicted values from the weak learners. The output of the meta-learner is the final predictions. Thus, the meta-learner is trained to integrate the predictions of any given base model optimally. Stacking aims to increase overall performance by mitigating the shortcomings of individual models.

We developed a stacking ensemble-based imputation method, namely S-UImpute, to impute missing data in the considered datasets. The main motivation for developing the ensemble imputation method is the inconsistent performance of existing imputation techniques across datasets. It was observed that a method that provided good forecasting performance on some of the considered datasets could not provide the same performance on the remaining datasets or in different missing data ratios. For this reason, we developed an ensemble model to achieve more consistent performance and combine the strengths of existing imputation methods, which have a positive impact on forecasting performance.

In the proposed method, S-UImpute, the weak learners are the BFill, Linear Interpolation, Kalman Filter and SVR, which outperform other imputation methods when tested individually on the considered datasets. Gradient Boosting Regression (GBR), which can better capture these complex relationships than a generalized linear model-based approach ([Zhang et al., 2019](#)), was selected as the meta-learner.

Gradient boosting is a popular machine learning algorithm that combines individual decision trees, acting as weak learners, to create a strong learner ([Park et al., 2023](#)).

Basically, Gradient boosting creates an ensemble where each weak learner attempts to correct the errors of its predecessors. The gradient-boosting algorithm implements an iterative process that calculates the gradient of the loss function at each iteration and tries to minimize this gradient.

For training set D defined earlier, the gradient boosting algorithm tries to find an approximation $\hat{F}(x)$ of the function $F^*(x)$ that maps instances to outputs by minimizing the expected value of a loss function $L(y, F(x))$ (Bentéjac, Csörgő & Martínez-Muñoz, 2021). The additive approximation of $F^*(x)$ is obtained iteratively by a weighted sum of the functions:

$$F_m(x) = F_{m-1}(x) + \rho_m h_m(x) \quad (7)$$

where ρ_m denotes the weight of the m th function $h_m(x)$ and $h_m(x)$ refers to the weak learners (Rizkallah, 2025).

A constant approximation for the first weak learner is obtained by:

$$F_0(x) = \arg \min_{\alpha} \sum_{i=1}^n L(y_i, \alpha). \quad (8)$$

Then the subsequent models are expected to minimize (Bentéjac, Csörgő & Martínez-Muñoz, 2021).

$$(\rho_m, h_m(x)) = \arg \min_{\rho, h} \sum_{i=1}^n L(y_i, F_{m-1}(x_i) + \rho h(x_i)). \quad (9)$$

Each of the functions $h_m(x)$ is considered as a step in a gradient descent optimization of $F^*(x)$ and trained on D . The pseudo-residuals are calculated as follows, where r_{mi} is:

$$r_{mi} = \left[\frac{\partial L(y_i, F(x))}{\partial F(x)} \right]_{F(x)=F_{m-1}(x)}. \quad (10)$$

Then, line search is applied to solve this optimization problem to determine ρ_m weight values.

The proposed stacking ensemble method, S-UImpute, is visualized in Fig. 5. After the train and test split, each of the weak learners is trained separately with the training data, thus ensuring that the training data is imputed with each weak learner. In this process, missing data in the training set are first imputed separately by using baseline imputation techniques: BFill, linear interpolation, Kalman Filter, and SVR. Let x_t denote the missing data for the time period t . Each weak learner predicts the missing value of x_t individually: $\hat{x}_t^{BFill}$, $\hat{x}_t^{LinearInterpolation}$, $\hat{x}_t^{Kalman}$, $\hat{x}_t^{SVR}$. These four outputs are then combined into a single feature vector:

$$z^t = [\hat{x}_t^{BFill}, \hat{x}_t^{LinearInterpolation}, \hat{x}_t^{Kalman}, \hat{x}_t^{SVR}]. \quad (11)$$

This vector is used as the input to a GBR, which acts as a meta-learner. The nonlinear mapping function of GBR, $f(\cdot)$, generates the final imputed value as:

$$\hat{x}_t = f(z_t). \quad (12)$$

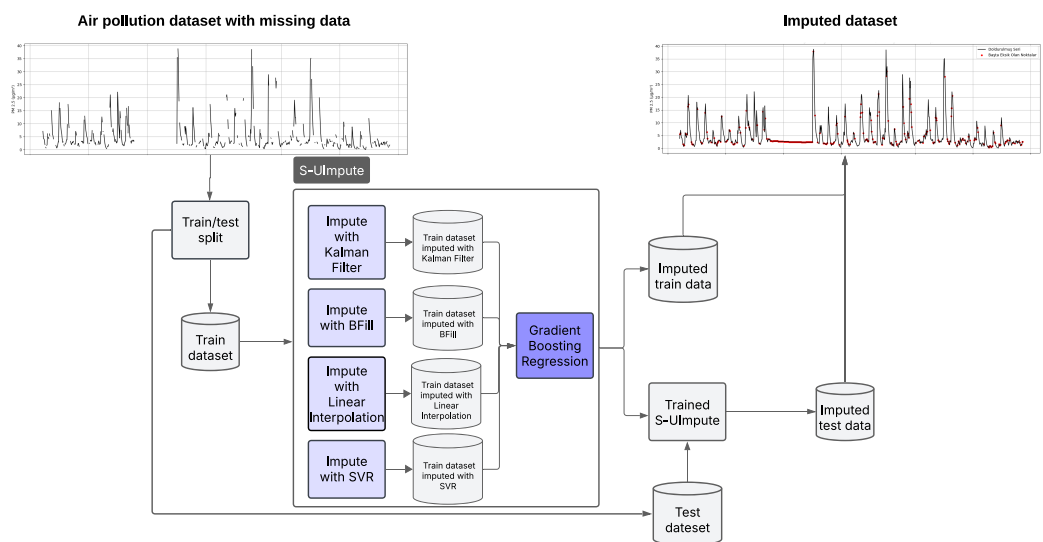


Figure 5 The proposed stacking ensemble imputation method, S-UImpute.

Full-size DOI: [10.7717/peerj-cs.3904/fig-5](https://doi.org/10.7717/peerj-cs.3904/fig-5)

In other words, the GBR model is trained using the outputs of the weak learners, and it generates the final imputed values. Concisely, the missing data in the training data is imputed with the trained meta-learner outputs.

Afterwards, the imputed training set and the original test set are combined to impute missing data in the test data. Currently, the dataset contains only the missing data in the test part, because all missing data in the training data have been filled previously. Individual imputation is performed for the test data using baseline imputation methods mentioned as weak learners previously trained on the training dataset. The resulting feature vector, z_t , is fed into the meta-learner previously trained on the training dataset as input, thus resulting in the final filled values for the test dataset.

During the imputation process, all original observed values are preserved, and only missing values are replaced. Since the test data is imputed after the training data has been imputed, there is no risk of data leakage. After imputation, prediction models are developed using the training and test datasets, which no longer contain missing values.

EXPERIMENTAL STUDY

Experimental settings

The performance of the proposed stacking ensemble model, S-UImpute, was evaluated based on the forecasting performance after imputing the data. Comparisons were made for the forecasts obtained for the original non-missing data. In other words, to ensure a fair comparison, forecasted values corresponding to imputed data were excluded from the performance metric calculations. In addition to the S-UImpute, the data were imputed using various existing methods for univariate time series, including ZeroFill, MeanFill, MedianFill, FFill, BFill, Interpolation, Kalman Filter, and SVR. Moreover, to evaluate the impact of imputation methods on forecasting performance, forecasts were conducted on the imputed datasets generated by each imputation method utilizing ARIMA, SVR, and LSTM models.

Table 2 Parameter search spaces.

Model	Parameter	Value
SVR	C	{1, 10, 100}
	ϵ	{0.01, 0.1}
	Kernel	Radial Basis Function (RBF)
	γ	{scale, 0.1}
LSTM	Number of hidden neurons	{10, 30, 50}
	Batch size	{1, 50, 100}
	Epoch	{50, 100}
ARIMA	p	{0, 1, 2, 3}
	d	{0, 1, 2}
	q	{0, 1, 2, 3}
GBR	Learning rate (α)	{0.05, 0.1, 0.2}
	max_depth	{3, 4, 5}

As the daily cycle was detected when the considered datasets were analyzed, the lag value was determined to be 24 for both the imputation method and the forecasting model utilizing SVR, and also the LSTM forecasting model. The dataset was first split into training (80%) and test (20%) sets. After the missing values were imputed for both sets, the data were normalized using the min–max method, with the scaler fitted on the training set and applied to the test set. Finally, the forecasting models were trained. Before calculating the evaluation metrics, the normalized predictions and target values were converted back to the original scale using the inverse min–max transformation.

To determine the parameters of the machine learning models, random search methodology was applied for LSTM and GBR model parameters, and grid search methodology was applied for SVR and ARIMA parameters. The parameter search spaces for each model are given in [Table 2](#).

The Kalman Filter was implemented using a standard random-walk state-space formulation with fixed covariance parameters and no hyperparameter tuning, as commonly adopted in time-series imputation studies. Since BFill and linear interpolation imputation methods are applied based on observed data itself, they do not require any parameter optimization. Therefore, no hyperparameter tuning was performed for these weak learners.

As given in [Table 2](#), hyperparameter tuning was conducted only for the learning-based components of the S-UImpute, namely SVR used as a weak learner and GBR as the meta-learner. The selected hyperparameters for these models are summarized in [Table 3](#).

All imputation and forecasting models were implemented in Python and executed on Google Colab Pro. The code is publicly available in our GitHub repository ([Zierzurum, 2025](#)).

Evaluation metrics

To measure the forecast errors, we use the Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and modified symmetric Mean Absolute Percentage Error

Table 3 Hyperparameters used in the S-UImpute.

Model	Hyperparameter	Value
SVR	Kernel	RBF
	C	C = 1
	ϵ	$\epsilon = 0.1$
	γ	scale
GBR	Number of estimators	100
	Learning rate (α)	0.1
	Max depth	3

(msMAPE) to avoid problems with dividing by zero or very close values (*Shcherbakov et al., 2013*). The formulations of these error measures are given in Eqs. (13)–(15).

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|. \quad (13)$$

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}. \quad (14)$$

$$msMAPE = \frac{1}{n} \sum_{i=1}^n \frac{|y_i - \hat{f}_i|}{\frac{|y_i| + |\hat{f}_i|}{2} + S_i} \quad (15)$$

$$S_i = \frac{1}{i-1} \sum_{k=1}^{i-1} |y_k - \bar{y}_{i-1}|, \quad \bar{y}_{i-1} = \frac{1}{i-1} \sum_{k=1}^{i-1} y_k.$$

Computational results

To evaluate the models, we used the msMAPE metric. To demonstrate the performance of S-UImpute, the imputation methods and forecasting models were run 10 times, and the average values were reported. The msMAPE values obtained as a result of the forecasts made for the test datasets with the LSTM, SVR, and ARIMA models are visualized in Figs. 6, 7, 8, respectively. Additionally, the MAE, RMSE, and msMAPE values obtained for each forecasting model are reported in Tables 4, 5, and 6.

Evaluating the LSTM model's forecasts, we can conclude that using the S-UImpute imputation method yields the best prediction performance in terms of msMAPE. The forecasting model trained on imputed data with S-UImpute provides more consistent, less fluctuating forecast performance as the rate of missing data increases. When the rate of missing data increases, it is seen that the MeanFill, MedianFill, and ZeroFill methods deteriorate test forecast performance. Although the performance of the FFill, BFill, InterpolationFill, Kalman Filter, and SVR imputation methods is generally comparable, it is worth noting that they occasionally exhibit variations at specific missing data rates. Using the S-UImpute imputation method with the LSTM model improves the msMAPE values obtained for the test dataset, resulting in more consistent forecast performance.

A similar pattern is observed when we compare the test performance obtained with SVR. For the test dataset, the SVR model trained with S-UImpute produced msMAPE

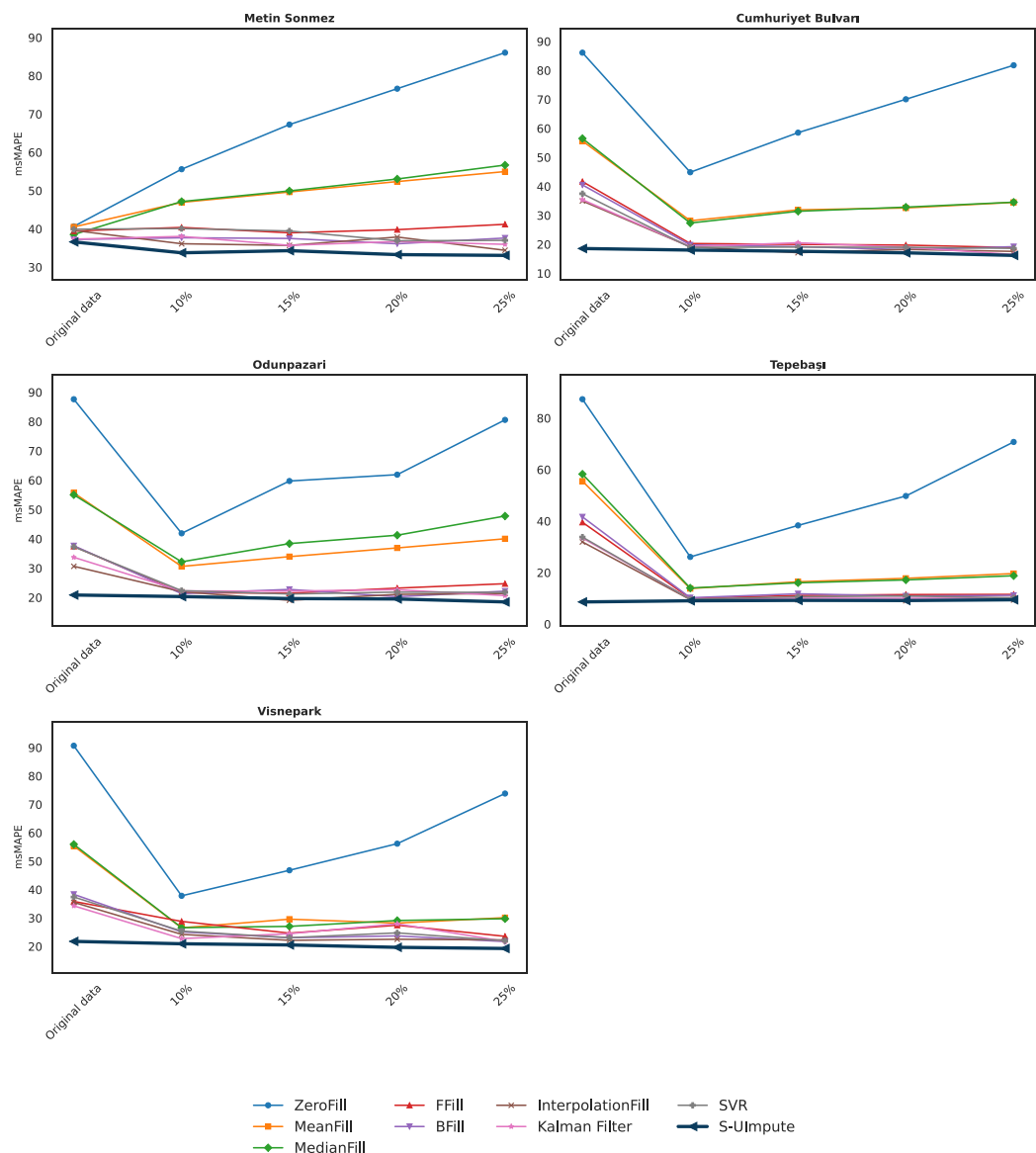


Figure 6 LSTM forecasting performance for different missing data rates (msMAPE values).

Full-size DOI: 10.7717/peerj-cs.3904/fig-6

values that were generally better than models trained with data imputed using other imputation methods. Furthermore, the msMAPE values obtained from the S-UImpute + SVR model were consistent and did not exhibit large variation across different missing data rates.

Lastly, regardless of the method used to fill in the missing data, the ARIMA model test results show that, even with parameter optimization, the forecasting performance is primarily in the “inaccurate forecast” category based on the MAPE scale (Lewis, 1982). It is not possible to conclude that a single imputation method is successful, even though its forecasting performance can only be categorized as “reasonable forecast” in the Tepebaşı

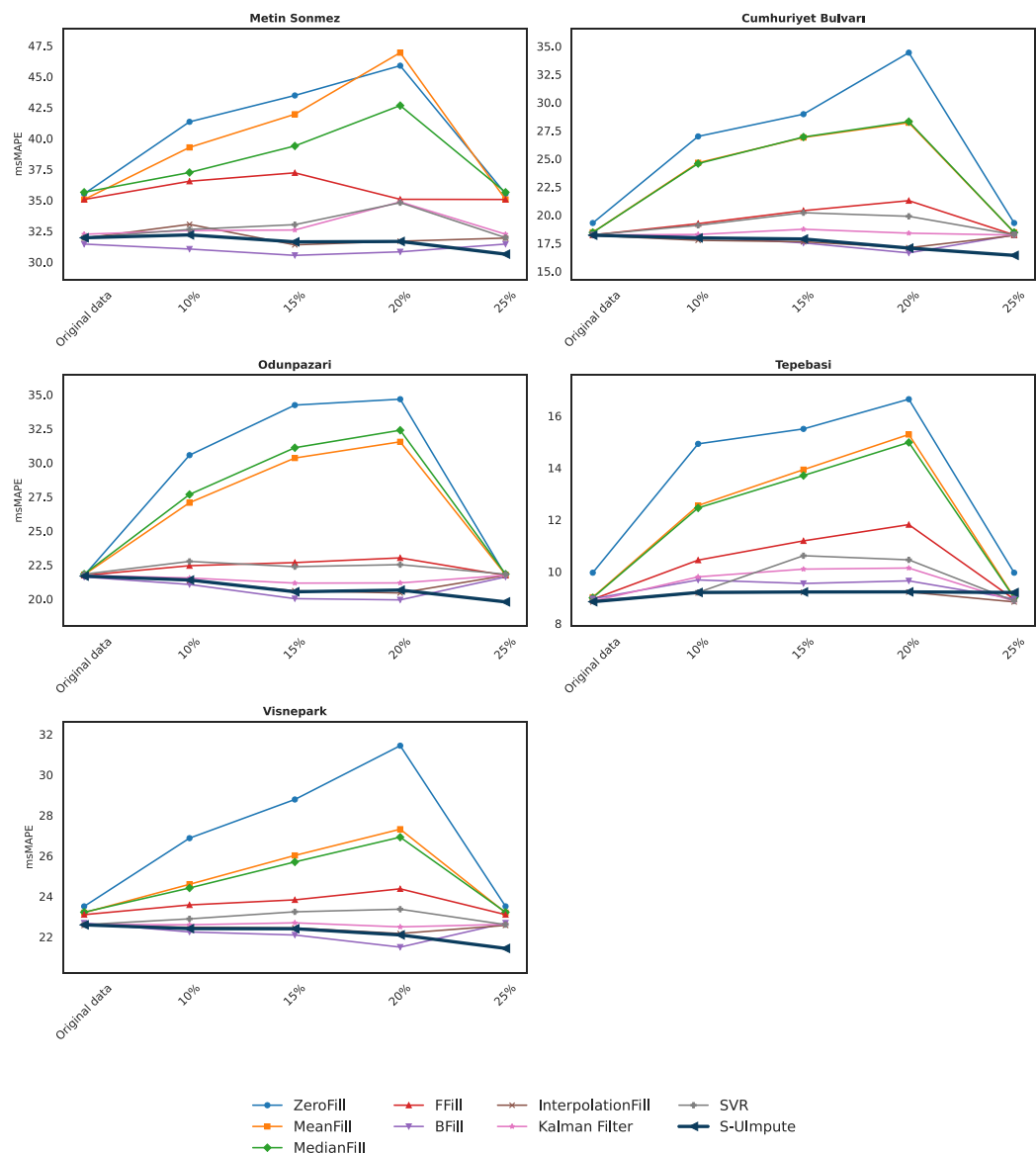


Figure 7 SVR forecasting performance for different missing data rates (msMAPE values).

Full-size DOI: 10.7717/peerj-cs.3904/fig-7

dataset. However, one could argue that S-UImpute performs better on the Tepebaşı dataset with a missing rate of 10%.

We also compared LSTM, SVR, and ARIMA models to assess their performance under different levels of missing data. LSTM generally outperformed SVR on the test datasets. In the Metin Sönmez and Cumhuriyet Bulvarı datasets, the SVR model outperformed LSTM with negligible msMAPE differences; however, LSTM dominated in terms of MAE and RMSE. Unfortunately, despite parameter optimization, the ARIMA model generated “inaccurate forecasts” based on the msMAPE scale in almost all of the datasets and at varying missing data rates. However, when the performance metrics obtained using the ARIMA model are compared with the LSTM and SVR models, it is seen that ARIMA

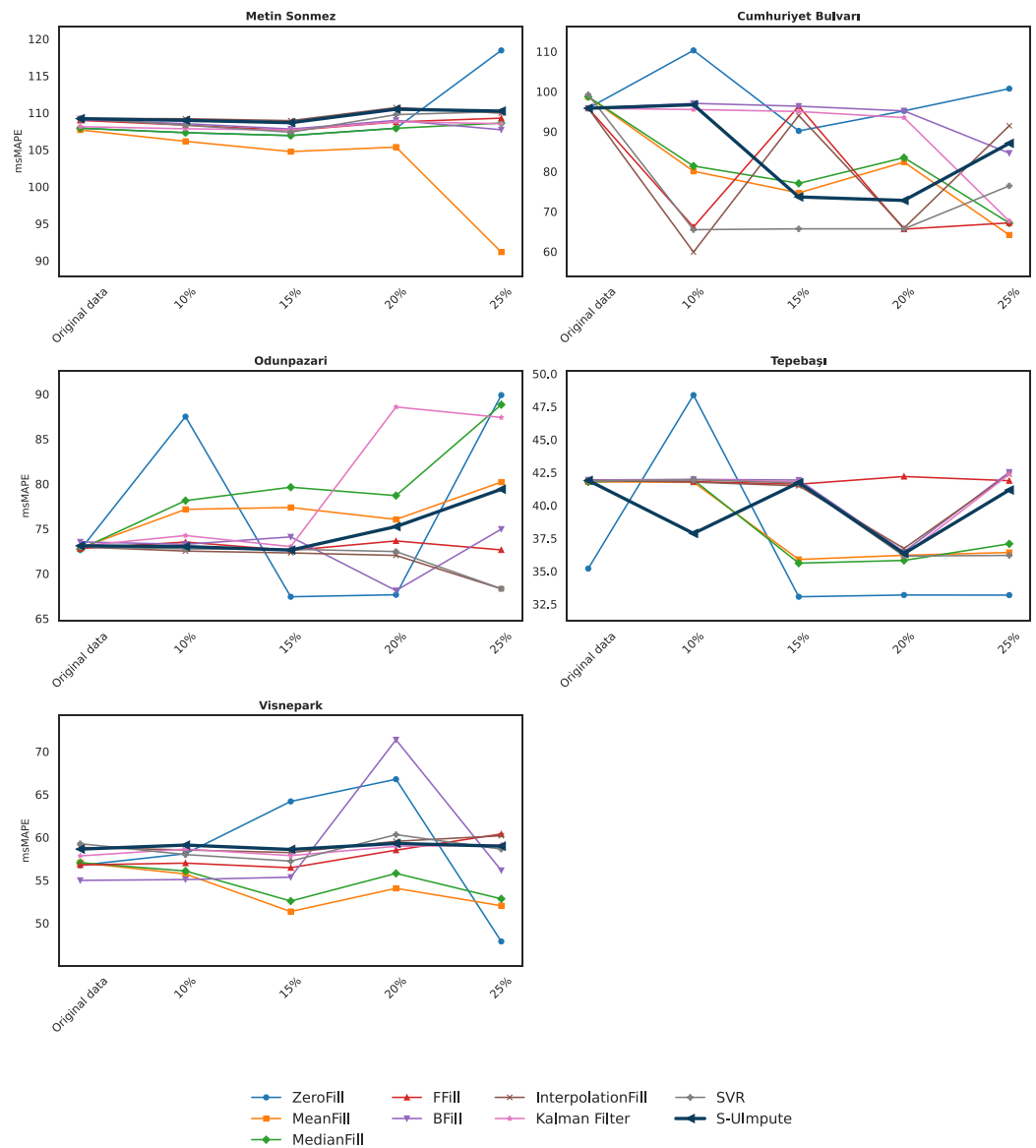


Figure 8 ARIMA forecasting performance for different missing data rates (msMAPE values).

Full-size DOI: [10.7717/peerj-cs.3904/fig-8](https://doi.org/10.7717/peerj-cs.3904/fig-8)

performs well behind SVR and LSTM. Given its poor forecasting performance on these datasets, one could argue that the developed ARIMA models are not suitable for this forecasting task.

Execution times

Figure 9 presents the average imputation time for each method. Among these methods, the proposed S-UImpute method has the highest average computation time. However, this corresponds to only approximately 5 s on average, which is acceptable for most practical applications, especially given that it leads to more accurate forecasts.

Table 4 LSTM model performance across different imputation methods and missing data rates.

Dataset	Method	Original data			10% Missing			15% Missing			20% Missing			25% Missing		
		MAE	RMSE	msMAPE	MAE	RMSE	msMAPE	MAE	RMSE	msMAPE	MAE	RMSE	msMAPE	MAE	RMSE	msMAPE
Metin Sönmez	ZeroFill	2.8730	5.7953	40.6564	3.6639	6.9246	55.5940	4.0516	7.4390	67.2617	4.5370	8.4673	76.6404	5.0192	8.8157	86.0514
	MeanFill	2.8021	5.6484	40.5118	3.3634	6.2545	46.9363	3.5663	6.5407	49.6425	3.9806	7.4741	52.3659	4.3702	7.8371	54.9785
	MedianFill	2.7300	5.8069	38.5346	3.4160	6.4711	47.1498	3.6618	6.8412	49.9259	4.1640	7.8622	53.0430	4.3809	8.2624	56.6791
	FFill	2.7075	5.1245	39.4591	2.7347	5.3374	40.3846	2.6805	4.9153	38.9784	2.7699	5.5470	39.8039	2.9565	5.4828	41.1997
	BFill	2.5588	5.0324	37.2289	2.6940	4.9907	37.6846	2.6209	4.8838	37.5036	2.5646	4.8439	36.1173	2.7564	5.0991	37.6147
	InterpolationFill	2.6816	4.9980	39.6001	2.4962	4.8792	36.1218	2.3902	4.5173	35.6925	2.5854	4.6050	37.8714	2.4096	4.3382	34.4095
	Kalman Filter	2.6032	5.1191	37.3228	2.7048	5.0687	38.0423	2.4824	4.6699	35.7046	2.6088	4.8579	36.7090	2.5451	4.6637	35.9678
	SVR	2.9051	5.6078	39.9242	2.7625	5.0957	40.0890	2.6489	4.8380	39.4546	2.7996	5.5109	36.9051	2.8708	5.5782	37.0023
	S-UImpute	2.4880	4.9291	36.6020	2.3566	4.7193	33.7537	2.2984	4.4301	34.3140	2.3856	4.7794	33.2810	2.3142	4.4955	33.0760
Cumhuriyet Bulvarı	ZeroFill	5.0060	8.8265	86.1319	8.4151	14.3054	44.8741	10.3534	16.3791	58.5349	11.6239	17.6216	70.0563	12.9979	19.1881	81.7841
	MeanFill	4.4106	7.7080	55.6556	6.8201	11.6481	28.1446	7.8946	13.0486	31.9046	8.0435	13.2429	32.5566	8.4475	13.3658	34.4846
	MedianFill	4.5684	8.3479	56.5239	6.7839	11.7455	27.3549	7.6564	12.9684	31.4141	8.1960	13.6398	32.8389	8.4977	13.7995	34.5404
	FFill	2.9125	5.4937	41.6426	4.3452	6.6726	20.2876	4.4350	7.2759	19.9811	4.4765	7.8785	19.7438	4.2331	6.9709	18.8958
	BFill	2.9304	5.2297	40.5065	4.3645	6.7751	19.7958	4.3649	7.0072	19.2327	4.1244	7.1162	18.1617	4.2690	7.1159	19.1970
	InterpolationFill	2.4479	4.4366	34.9608	4.1128	6.1621	19.1518	3.8266	6.1017	17.1212	3.8933	5.7998	18.2604	3.6940	5.4800	17.5693
	Kalman Filter	2.5719	4.7349	35.3955	4.1676	6.4216	19.2526	4.4415	6.5789	20.5452	4.1541	6.3379	19.0150	3.9522	6.3833	16.4857
	SVR	2.8702	5.5686	37.4724	4.3454	7.2127	18.9836	4.7462	8.5554	19.0577	4.5788	8.0815	18.9800	4.4414	7.8303	18.6170
	S-UImpute	3.9657	6.1234	18.5600	4.0029	6.3146	18.0260	4.0284	6.4764	17.6150	3.8158	6.2053	17.0740	3.6687	6.1045	16.1880
Odunpazarı	ZeroFill	5.2626	9.0905	87.6444	5.8551	9.9392	41.8997	7.5955	12.2953	59.7411	7.8062	12.2198	61.9120	9.4447	14.2996	80.5952
	MeanFill	4.3769	7.9933	55.7011	5.1269	8.8464	30.6185	5.9726	10.5331	33.9538	6.2301	10.1604	36.9269	7.0397	11.5210	40.0435
	MedianFill	4.3692	8.2551	55.0531	5.4451	9.3420	32.1812	6.5955	11.2841	38.3997	6.9667	11.0861	41.2642	8.0129	12.6494	47.8034
	FFill	2.7217	5.5253	37.4668	3.7832	6.2483	21.6745	3.4697	5.6981	21.4879	3.7087	5.9146	23.2162	3.8336	5.8585	24.7212
	BFill	2.7539	5.1624	37.6598	3.6889	6.0268	21.3334	3.5755	5.3815	22.7804	3.4465	5.7382	20.2799	3.5823	5.7449	22.1836
	InterpolationFill	2.2785	4.4897	30.6673	3.4653	5.4318	22.0170	3.1818	5.0383	19.0534	3.3001	5.2438	21.0875	3.2351	4.9851	21.3798
	Kalman Filter	2.5295	4.7978	33.7289	3.5944	5.5096	22.2522	3.4881	5.1324	22.1941	3.5651	5.3631	22.6821	3.3121	5.2443	20.6608
	SVR	2.9028	5.5197	37.3129	3.7667	6.4460	22.3554	3.8501	7.0931	21.2207	3.8098	6.6826	21.8416	4.0148	7.6067	21.7597
	S-UImpute	3.4365	5.5175	20.8660	3.5624	5.6305	20.3380	3.2980	5.2549	19.6560	3.5104	5.6461	19.5150	3.2888	5.6631	18.5060
Tepebaşı	ZeroFill	5.0088	8.9265	87.3639	8.0421	13.8189	26.1029	10.7566	16.9233	38.3394	13.0493	18.7597	49.7696	16.5873	21.8014	70.7609
	MeanFill	4.3243	7.7536	55.5431	5.8937	8.9925	13.8202	6.9384	10.2504	16.4619	7.5239	10.6144	17.7874	8.3130	11.3319	19.6400
	MedianFill	4.4440	8.4174	58.2769	5.9537	8.9340	14.0234	6.8000	10.1005	16.0460	7.2549	10.4536	17.1671	7.9455	11.1015	18.7913
	FFill	2.8464	5.6855	39.5559	4.2559	6.2948	10.0011	4.6740	6.6859	11.1418	4.8235	6.9614	11.4822	4.8970	7.1514	11.5420
	BFill	2.8130	5.3820	41.6479	4.3557	6.2404	10.2593	4.9158	6.6478	11.7791	4.6401	6.9305	10.9351	4.7411	6.9184	11.1837
	InterpolationFill	2.3409	4.4338	31.9168	4.0006	5.5663	9.5324	3.9973	5.7972	9.4454	3.9003	5.7102	9.1249	3.9071	5.4761	9.3851
	Kalman Filter	2.5220	4.6810	33.4449	4.1657	6.0407	9.8066	4.3092	6.1308	10.0997	4.2970	6.1721	10.0299	4.3329	6.2427	10.1998
	SVR	2.8025	5.6091	33.8156	4.0875	5.9392	9.6702	4.4051	6.4963	10.6314	4.5778	6.3244	10.9351	4.3275	6.3957	10.2271
	S-UImpute	3.6198	5.2267	8.6180	3.8130	5.4749	9.0667	3.9171	5.6797	9.2255	3.8341	5.5338	9.1640	3.9863	5.7557	9.4538
Vişnepark	ZeroFill	5.0379	9.0735	90.6663	12.4985	20.2770	37.8280	13.7081	22.4653	46.8134	15.2778	23.7368	56.1858	17.8953	26.3092	73.8394
	MeanFill	4.4290	7.8401	55.3701	9.9274	16.5155	26.5490	11.2623	17.7171	29.5952	10.6576	17.4735	28.2194	11.4838	17.9730	30.1301
	MedianFill	4.4389	8.3469	55.9371	10.0277	16.7578	26.6019	10.3356	17.6806	27.0602	11.0012	17.6657	29.1292	11.3368	18.1839	29.7621
	FFill	2.6508	5.5608	35.8436	11.1185	16.6215	28.8097	9.3745	15.5525	24.6813	9.0733	15.6448	27.4783	8.9981	14.6222	23.5541
	BFill	2.7287	5.2475	38.3447	9.0206	15.6420	25.1152	8.7731	15.4935	23.0651	8.7916	15.3788	23.7086	8.4986	15.6895	21.7101
	InterpolationFill	2.4353	4.3930	35.5238	9.2866	14.9192	24.2514	8.4679	14.8857	22.1493	8.7917	15.0038	22.5276	8.2938	13.5757	22.3043
	Kalman Filter	2.5411	4.7777	34.2966	8.6321	14.4070	22.7450	8.8290	15.3835	24.4813	9.0326	15.4194	27.9611	8.4694	13.4969	21.7541
	SVR	2.8851	5.4905	37.4085	9.7850	15.6826	25.3990	8.5726	15.4801	23.0941	9.7670	16.2450	24.7798	8.4473	14.0241	21.9838
	S-UImpute	8.2895	14.4429	21.7819	7.9493	14.1608	20.9410	7.7762	14.0789	20.5251	7.4506	13.9816	19.6846	7.2123	12.8178	19.2857

Note:
The bold entries represent the best-performing results for each metric, the minimum MAE, RMSE, and msMAPE values obtained across all models.

Table 5 SVR model performance across different imputation methods and missing data rates.

Dataset	Method	Original data			10% Missing			15% Missing			20% Missing			25% Missing		
		MAE	RMSE	msMAPE	MAE	RMSE	msMAPE	MAE	RMSE	msMAPE	MAE	RMSE	msMAPE	MAE	RMSE	msMAPE
Metin sönmez	ZeroFill	2.6623	5.8867	35.5209	3.1147	6.8208	41.3441	3.1405	6.4300	43.4773	3.5067	7.5264	45.8847	2.6623	5.8867	35.5209
	MeanFill	2.5893	5.7536	35.0619	2.9139	6.0767	39.2736	3.1131	6.2466	41.9576	3.5240	7.1298	46.9393	2.5893	5.7536	35.0619
	MedianFill	2.6557	5.8408	35.6438	2.8396	6.2240	37.2423	3.0898	6.6206	39.3909	3.3715	7.3348	42.6625	2.6557	5.8408	35.6438
	FFill	2.5633	5.4119	35.0619	2.6875	5.5617	36.5369	2.7198	5.5693	37.2166	2.7880	6.1001	35.0746	2.5633	5.4119	35.0619
	BFill	2.3733	5.1035	31.4519	2.3489	5.0822	31.0494	2.2504	4.8232	30.5375	2.3391	4.9928	30.8254	2.3733	5.1035	31.4519
	InterpolationFill	2.4163	5.1660	31.9369	2.4459	5.1468	33.0489	2.3357	4.9379	31.3904	2.4256	5.0553	31.6850	2.4163	5.1660	31.9369
	Kalman Filter	2.4364	5.2163	32.2508	2.4558	5.2391	32.5415	2.4473	5.1612	32.5916	2.6454	5.4057	34.8682	2.4364	5.2163	32.2508
	SVR	2.4351	5.2789	31.9706	2.5001	5.3115	32.6389	2.5358	5.3509	33.0239	2.7408	5.8004	34.8003	2.4351	5.2789	31.9706
	S-UImpute	2.4064	5.1762	31.9646	2.4234	5.1575	32.1947	2.3674	5.0008	31.6295	2.4424	5.1471	31.6647	2.4523	5.0692	30.6360
Cumhuriyet bulvarı	ZeroFill	4.3375	7.0000	19.2885	6.1397	9.6275	26.9884	6.7259	10.8273	28.9651	8.2231	13.4758	34.4330	4.3375	7.0000	19.2885
	MeanFill	4.2428	6.8186	18.4561	5.8080	9.7969	24.6437	6.4358	11.1488	26.8915	6.7960	11.7106	28.2055	4.2428	6.8186	18.4561
	MedianFill	4.2396	6.8209	18.4475	5.8317	10.1545	24.5717	6.4798	11.3038	26.9343	6.8834	11.9771	28.3050	4.2396	6.8209	18.4475
	FFill	4.1779	6.7223	18.2288	4.4453	7.1660	19.2319	4.8219	8.0010	20.3877	5.0077	8.5978	21.2753	4.1779	6.7223	18.2289
	BFill	4.1773	6.7180	18.2252	4.2154	6.8943	18.0649	3.9707	6.5579	17.5194	3.7058	6.0229	16.6329	4.1773	6.7180	18.2252
	InterpolationFill	4.1591	6.7062	18.1813	4.0580	6.5411	17.7505	4.0093	6.4935	17.6233	3.8771	6.1139	17.0874	4.1591	6.7062	18.1813
	Kalman Filter	4.1771	6.7171	18.2162	4.2048	6.7844	18.2706	4.3250	6.9274	18.7346	4.3017	6.7944	18.3836	4.1771	6.7171	18.2162
	SVR	4.1818	6.7250	18.2435	4.4681	7.3630	19.0814	4.7673	8.1455	20.2041	4.6720	8.0766	19.8837	4.1818	6.7250	18.2435
	S-UImpute	4.1732	6.7155	18.2019	4.1568	6.7193	17.9619	4.1255	6.7479	17.8674	3.9293	6.3711	17.0592	3.7972	6.3209	16.4098
Odunpazarı	ZeroFill	3.5577	5.8114	21.7348	4.7359	8.5790	30.5573	5.5073	9.8818	34.2316	5.7672	9.6430	34.6651	3.5577	5.8114	21.7348
	MeanFill	3.5431	5.7654	21.7879	4.4566	8.1765	27.0686	5.1751	9.6305	30.3482	5.3676	9.5378	31.5364	3.5431	5.7654	21.7879
	MedianFill	3.5523	5.7862	21.8304	4.5177	8.2773	27.6741	5.2778	9.8540	31.1044	5.5049	9.6733	32.3858	3.5523	5.7862	21.8304
	FFill	3.5296	5.7224	21.7363	3.7053	6.0482	22.4476	3.6895	6.0739	22.6762	3.8670	6.2437	23.0209	3.5296	5.7224	21.7363
	BFill	3.5317	5.7057	21.6223	3.4498	5.6361	21.0719	3.1799	5.0628	20.0290	3.2748	5.4826	19.9396	3.5317	5.7057	21.6223
	InterpolationFill	3.5338	5.7096	21.7317	3.4693	5.6536	21.3561	3.2990	5.2083	20.6061	3.3367	5.5478	20.4459	3.5338	5.7096	21.7317
	Kalman Filter	3.5399	5.7160	21.7439	3.5139	5.6725	21.5474	3.4149	5.3045	21.1692	3.4798	5.6616	21.1829	3.5399	5.7160	21.7439
	SVR	3.5400	5.7153	21.8249	3.7407	6.3550	22.7619	3.7766	6.5483	22.3772	3.8250	6.6916	22.5246	3.5400	5.7153	21.8249
	S-UImpute	3.5344	5.7086	21.6841	3.4987	5.7202	21.3983	3.3534	5.3733	20.5334	3.4500	5.7310	20.6587	3.3231	5.6478	19.7917
Tepebaşı	ZeroFill	4.1239	6.2484	9.9597	6.0374	9.3580	14.9256	6.4015	9.7611	15.5008	6.8786	10.5019	16.6469	4.1239	6.2484	9.9597
	MeanFill	3.7553	5.3117	9.0133	5.3692	8.4804	12.5508	5.9855	9.3881	13.9310	6.4583	9.6098	15.2861	3.7553	5.3117	9.0133
	MedianFill	3.7501	5.3248	8.9932	5.3155	8.4622	12.4563	5.8789	9.3120	13.7050	6.3277	9.5216	14.9799	3.7501	5.3248	8.9932
	FFill	3.7199	5.2958	8.9360	4.4294	6.4223	10.4450	4.7919	7.1124	11.1873	5.0707	7.5395	11.8120	3.7199	5.2958	8.9360
	BFill	3.7400	5.3569	8.9540	4.0966	5.8573	9.6804	4.0718	5.8617	9.5447	4.1113	5.9872	9.6462	3.7400	5.3569	8.9540
	InterpolationFill	3.6814	5.2167	8.8350	3.8756	5.4918	9.1820	3.9087	5.5927	9.2037	3.8973	5.5937	9.2157	3.6814	5.2167	8.8350
	Kalman Filter	3.6938	5.2360	8.8615	4.1606	5.9562	9.8000	4.3171	6.3072	10.0978	4.3215	6.2243	10.1397	3.6938	5.2360	8.8615
	SVR	3.6974	5.2379	8.8715	3.8795	5.5051	9.1993	4.5002	6.6932	10.6154	4.4280	6.3800	10.4564	3.6974	5.2379	8.8715
	S-UImpute	3.6883	5.2509	8.8422	3.8795	5.5051	9.1993	3.9164	5.6580	9.2165	3.8781	5.5553	9.2212	3.8761	5.6422	9.1965
Vişnepark	ZeroFill	8.7394	15.2156	23.5080	9.8274	17.2115	26.8743	10.8365	18.6488	28.7823	11.6592	19.8003	31.4372	8.7394	15.2156	23.5080
	MeanFill	8.6770	14.9962	23.1976	9.1201	15.6354	24.5956	9.7426	16.7314	26.0205	10.1831	17.1913	27.3155	8.6770	14.9962	23.1976
	MedianFill	8.6830	15.0154	23.2202	9.0765	15.7168	24.4150	9.6603	16.7866	25.7001	10.0837	17.2299	26.9231	8.6830	15.0154	23.2202
	FFill	8.6617	14.9841	23.1000	8.8492	15.3800	23.5789	8.9993	15.6263	23.8293	9.1806	15.9797	24.3734	8.6617	14.9841	23.1000
	BFill	8.4876	14.8420	22.6843	8.1283	14.1295	22.2447	8.1682	14.6713	22.0945	7.7768	14.0501	21.4952	8.4876	14.8420	22.6843
	InterpolationFill	8.4281	14.6409	22.5699	8.3122	14.2431	22.4752	8.3454	14.5733	22.4412	8.1659	14.3298	22.1683	8.4281	14.6409	22.5699
	Kalman Filter	8.4531	14.6474	22.6202	8.3757	14.3400	22.5990	8.5019	14.8339	22.6948	8.2999	14.4691	22.4945	8.4531	14.6474	22.6202
	SVR	8.4432	14.6151	22.6062	8.4805	14.6592	22.8910	8.6743	15.1106	23.2394	8.6242	14.9577	23.3676	8.4432	14.6151	22.6062
	S-UImpute	8.4245	14.5844	22.5953	8.2370	14.1262	22.4090	8.3146	14.5548	22.4036	8.0765	14.2280	22.0983	8.0097	13.3903	21.4327

Note:
The bold entries represent the best-performing results for each metric, the minimum MAE, RMSE, and msMAPE values obtained across all models.

Table 6 ARIMA model performance across different imputation methods and missing data rates.

Dataset	Method	Original data			10% Missing			15% Missing			20% Missing			25% Missing		
		MAE	RMSE	msMAPE	MAE	RMSE	msMAPE	MAE	RMSE	msMAPE	MAE	RMSE	msMAPE	MAE	RMSE	msMAPE
Metin sönmez	ZeroFill	8.2810	14.3972	107.9128	8.2840	14.5020	107.3141	8.1892	14.2876	106.9223	8.3297	14.6371	107.9075	8.7338	14.8824	118.4455
	MeanFill	8.2746	14.3918	107.6840	8.2508	14.4739	106.1405	8.1271	14.2343	104.7518	8.2577	14.5779	105.3589	7.8448	14.0160	91.1550
	MedianFill	8.2813	14.3973	107.9205	8.2840	14.5020	107.3141	8.1892	14.2876	106.9223	8.3297	14.6371	107.9075	8.4819	14.6822	108.6162
	FFill	8.3104	14.4215	108.9790	8.3124	14.5256	108.3387	8.2083	14.3037	107.6086	8.3532	14.6561	108.7646	8.5007	14.6979	109.2791
	BFill	8.3132	14.4238	109.0809	8.3177	14.5300	108.5341	8.2138	14.3082	107.8069	8.3590	14.6608	108.9793	8.4558	14.6601	107.7082
	InterpolationFill	8.3174	14.4272	109.2384	8.3357	14.5447	109.2013	8.2446	14.3338	108.9440	8.4047	14.6970	110.7099	8.5196	14.7136	109.9567
	Kalman Filter	8.2871	14.4022	108.1290	8.2987	14.5143	107.8409	8.2100	14.3051	107.6700	8.3542	14.6570	108.8033	8.4794	14.6801	108.5278
	SVR	8.3172	14.4270	109.2278	8.3120	14.5252	108.3226	8.2018	14.2982	107.3749	8.3797	14.6772	109.7538	8.5254	14.7184	110.1691
	S-UImpute	8.3166	14.4266	109.2087	8.3298	14.5399	108.9823	8.2370	14.3276	108.6639	8.3990	14.6924	110.4884	8.5271	14.7197	110.2349
Cumhuriyet bulvarı	ZeroFill	21.4503	30.8892	95.8223	22.9337	32.1678	110.2794	20.6824	29.9900	90.1610	21.4560	31.0556	95.1637	22.1612	31.5417	100.7171
	MeanFill	21.7778	31.1846	98.6677	19.3599	28.9009	80.0848	18.5225	27.8718	74.6646	19.8275	29.5350	82.3552	16.8538	25.6969	64.1392
	MedianFill	21.7811	31.1876	98.6970	19.5435	29.0811	81.3847	18.8877	28.2557	77.0507	19.9815	29.6836	83.4794	17.4243	26.6054	67.1401
	FFill	21.4503	30.8892	95.8223	17.1256	26.3849	66.1464	21.4337	30.6836	96.3399	17.1431	26.4229	65.5985	17.4320	26.6309	67.1737
	BFill	21.4503	30.8892	95.8223	21.5294	30.9378	97.0442	21.4337	30.6836	96.3399	21.4560	31.0556	95.1637	20.1731	29.7129	84.6068
	InterpolationFill	21.4503	30.8892	95.8223	15.9172	24.0567	59.8745	21.1553	30.4260	94.0069	17.1971	26.5092	65.8796	21.0621	30.5415	91.4460
	Kalman Filter	21.4503	30.8892	95.8223	21.3496	30.7749	95.4967	21.2773	30.5402	95.0179	21.2557	30.8721	93.4778	17.5105	26.7450	67.5992
	SVR	21.8390	31.2396	99.2090	17.0006	26.2063	65.4739	16.9970	26.0218	65.6801	17.1600	26.4540	65.6835	18.9898	28.5249	76.4259
	S-UImpute	21.4503	30.8892	95.8223	21.4634	30.8673	96.6957	18.1113	27.1845	73.6541	18.1812	27.5570	72.7812	20.4764	29.9844	87.0522
Odunpazarı	ZeroFill	13.6775	20.0410	72.6526	15.3801	22.1278	87.4958	12.9929	18.8073	67.4308	12.9673	18.5997	67.6529	15.5988	22.4715	89.8939
	MeanFill	13.7040	20.0848	72.8491	14.2231	20.7517	77.1662	14.3167	21.0085	77.3774	14.1057	20.6332	76.0428	14.5786	21.3437	80.2014
	MedianFill	13.6881	20.0586	72.7315	14.3423	20.9122	78.1382	14.5853	21.3481	79.6337	14.4333	21.0778	78.6982	15.4944	22.3636	88.8291
	FFill	13.6973	20.0738	72.7996	13.7567	20.0679	73.5257	13.7026	20.1166	72.5883	13.7942	20.1666	73.6466	13.6401	20.0376	72.6487
	BFill	13.7962	20.2345	73.5377	13.7233	20.0148	73.2741	13.9021	20.4237	74.0977	13.0312	18.7576	68.1172	13.9448	20.5109	74.9614
	InterpolationFill	13.7208	20.1125	72.9743	13.6219	19.8501	72.5173	13.6655	20.0574	72.3116	13.5766	19.8088	72.0357	13.0410	18.9447	68.3153
	Kalman Filter	13.7362	20.1379	73.0888	13.8523	20.2171	74.2504	13.7592	20.2057	73.0125	15.5069	22.2974	88.5724	15.3514	22.2106	87.4018
	SVR	13.7211	20.1129	72.9761	13.6548	19.9036	72.7622	13.7199	20.1438	72.7174	13.6355	19.9080	72.4679	13.0430	18.9493	68.3298
	S-UImpute	13.7385	20.1417	73.1062	13.6894	19.9598	73.0206	13.7078	20.1248	72.6274	13.9518	20.2820	75.2492	15.4158	22.5341	79.4060
Tepebaşı	ZeroFill	15.1108	19.0553	35.2025	18.9973	26.0024	48.3803	14.1062	18.8767	33.0630	14.2278	19.6363	33.1964	14.1908	20.0183	33.1861
	MeanFill	17.1104	24.1642	41.7871	17.0860	24.0904	41.7697	15.4760	19.2100	35.8943	15.6703	19.4378	36.2119	15.3760	22.0903	36.4238
	MedianFill	17.1117	24.1656	41.7914	17.1386	24.1464	41.9428	15.3355	19.1206	35.6150	15.4743	19.3195	35.8214	15.5995	22.3856	37.0831
	FFill	17.1430	24.1987	41.8943	17.0986	24.1042	41.8109	17.0341	24.0237	41.6141	17.3062	24.4404	42.1969	17.1271	24.1548	41.8802
	BFill	17.1557	24.2121	41.9360	17.1494	24.1582	41.9780	17.1296	24.1260	41.9281	15.8325	19.5868	36.4755	17.3232	24.3595	42.5303
	InterpolationFill	17.1438	24.1994	41.8968	17.0869	24.0916	41.7727	16.9955	23.9823	41.4878	15.9118	19.5653	36.7289	17.2754	24.3109	42.3709
	Kalman Filter	17.1357	24.1908	41.8701	17.1434	24.1519	41.9584	17.0906	24.0841	41.7997	15.6854	19.4737	36.1999	17.2759	24.3105	42.3730
	SVR	17.1354	24.1906	41.8691	17.1416	24.1499	41.9525	17.0628	24.0541	41.7088	15.6608	19.4569	36.1531	15.6291	19.3022	36.2033
	S-UImpute	17.1443	24.2000	41.8985	16.0507	20.9323	37.8721	17.0696	24.0617	41.7310	15.7565	19.5189	36.3459	16.9359	23.6243	41.1705
Vişnepark	ZeroFill	20.9268	32.6339	56.7443	21.1750	32.8583	58.0900	22.6884	34.6744	64.1916	23.0740	34.7899	66.7751	18.4851	29.2104	47.8669
	MeanFill	21.0041	32.7163	57.0491	20.5805	32.2348	55.7306	19.4877	31.3271	51.3502	20.0745	31.8361	54.0569	19.7281	31.0214	52.0294
	MedianFill	21.0026	32.7148	57.0432	20.6691	32.3275	56.0773	19.8289	31.7454	52.5726	20.5252	32.3206	55.8000	19.9552	31.3058	52.8344
	FFill	20.9382	32.6460	56.7891	20.9024	32.5749	56.9964	20.8518	32.8701	56.4563	21.1978	33.0168	58.5073	21.9091	33.4370	60.3911
	BFill	20.4742	32.1511	54.9853	20.4118	32.0524	55.0798	20.5706	32.5753	55.3572	24.0267	35.6356	71.3770	20.8448	32.3167	56.1460
	InterpolationFill	21.4448	33.1699	58.8265	21.2826	32.9656	58.5297	21.2935	33.3283	58.2248	21.4467	33.2675	59.5410	21.8600	33.3861	60.1892
	Kalman Filter	21.2022	32.9263	57.8385	21.3084	32.9914	58.6354	21.2063	33.2414	57.8694	21.3058	33.1260	58.9538	21.5932	33.1135	59.0978
	SVR	21.5504	33.2765	59.2604	21.1517	32.8352	57.9952	21.0476	33.0789	57.2309	21.6287	33.4416	60.3137	21.4765	32.9956	58.6247
	S-UImpute	21.3963	33.1209	58.6284	21.4219	33.1062	59.1016	21.3824	33.4181	58.5879	21.3867	33.2075	59.2904	21.5623	33.0819	58.9731

Note:
The bold entries represent the best-performing results for each metric, the minimum MAE, RMSE, and msMAPE values obtained across all models.

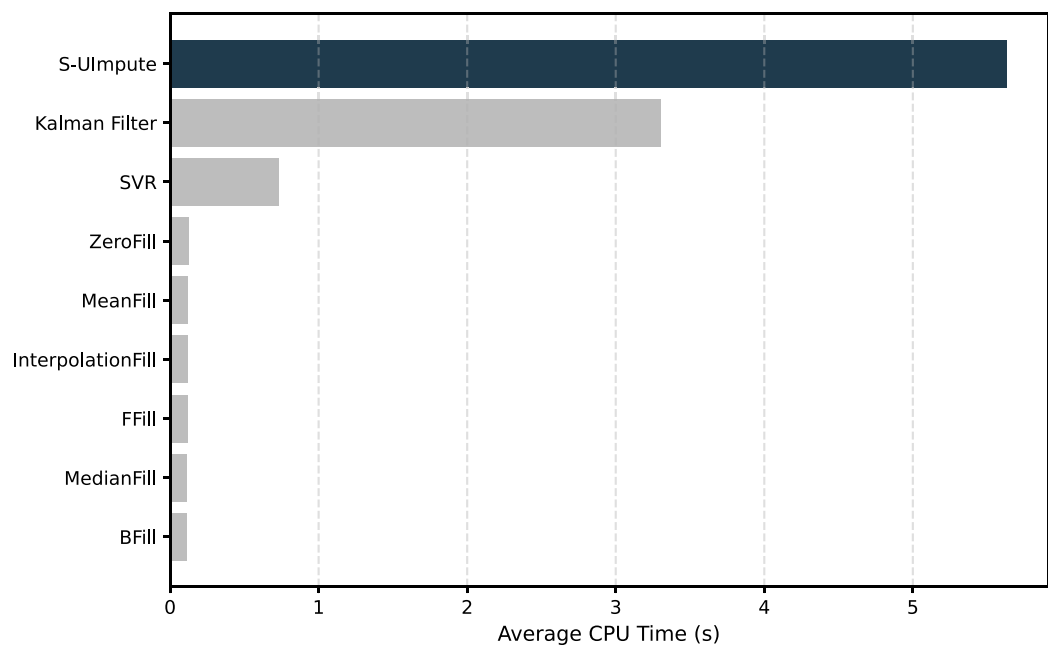


Figure 9 Imputation time.

Full-size DOI: 10.7717/peerj-cs.3904/fig-9

Generalization of model performance using cross-validation

A cross-validation experiment was conducted to accurately evaluate the model performance. Since random cross-validation is not appropriate for time-dependent data (Bergmeir & Benítez, 2012), cross-validation was performed using an expanding window strategy, following the evaluation framework recommended by Hyndman & Athanasopoulos (2021). In this strategy, the test dataset size is kept constant, the test dataset is sequentially moved into the training set, and the forecast origin is changed accordingly (Kingphai & Moshfeghi, 2023). Thus, the training set size increases with each fold, while the test set size remains constant, with only the forecast origin being shifted. Because it outperformed the other models tested, 5-fold cross-validation experiments were conducted using the LSTM model.

Table 7 reports the average performance metrics and standard deviations (shown in parentheses) obtained with 5-fold cross-validation for each missing data imputation method. For example, for the Metin Sönmez station, with a 25% missing data rate, S-UImpute achieved a mean MAE of 1.68 after 5-fold cross-validation. The standard deviation value after 5-fold cross-validation is 0.45.

Overall, the results show that S-UImpute gives lower MAE, RMSE, and msMAPE values for the LSTM model compared to other missing data imputation methods, both in the original data and as the missing data rate increases. In contrast, simple imputation methods such as ZeroFill, MeanFill, and MedianFill show higher errors as the rate of missing data increases. While methods such as Kalman filtering and interpolation outperform the simple imputation methods, they fall behind S-UImpute. In addition to having the lowest average errors, S-UImpute generally shows smaller standard deviation

Table 7 5-fold cross-validation performance metrics for the LSTM model (values in parentheses represent the standard deviation).

Original data		10% Missing					15% Missing					20% Missing					25% Missing				
		MAE	RMSE	msMAPE	MAE	RMSE	msMAPE	MAE	RMSE	msMAPE	MAE	RMSE	msMAPE	MAE	RMSE	msMAPE	MAE	RMSE	msMAPE		
Metin Sönmez	ZeroFill	3.46 (1.01)	5.27 (1.38)	58.20 (8.99)	7.58 (3.11)	11.52 (5.43)	37.00 (8.11)	7.39 (2.91)	11.18 (5.20)	36.40 (7.25)	7.30 (2.83)	11.14 (5.20)	35.91 (7.30)	7.63 (2.92)	11.49 (5.22)	37.47 (7.04)					
	MeanFill	3.61 (1.14)	5.30 (1.35)	60.25 (10.30)	7.70 (2.53)	10.70 (4.57)	38.30 (6.42)	7.87 (3.18)	10.89 (4.99)	38.55 (9.06)	8.04 (3.30)	11.19 (5.22)	39.10 (9.60)	7.84 (2.83)	10.72 (4.57)	38.81 (8.46)					
	MedianFill	3.89 (1.29)	5.56 (1.42)	63.52 (17.33)	7.28 (2.26)	10.37 (4.20)	36.54 (5.83)	7.27 (2.42)	10.31 (4.21)	36.52 (7.97)	7.88 (2.80)	10.91 (4.70)	38.89 (7.53)	7.07 (2.50)	9.98 (4.26)	35.64 (7.88)					
	FFill	3.69 (1.10)	5.34 (1.37)	62.11 (11.22)	6.66 (2.15)	9.57 (3.69)	33.71 (6.50)	6.63 (2.28)	9.18 (3.43)	33.76 (7.34)	6.46 (2.17)	9.16 (3.39)	32.91 (7.42)	6.66 (2.21)	9.54 (3.91)	33.59 (6.47)					
	BFill	3.43 (0.66)	5.10 (1.04)	59.05 (5.61)	6.16 (2.04)	8.54 (3.10)	32.07 (6.18)	5.88 (1.65)	8.37 (2.89)	30.63 (4.69)	6.47 (2.11)	9.27 (3.64)	32.84 (6.05)	6.39 (2.29)	8.98 (3.45)	32.67 (7.03)					
	InterpolationFill	3.87 (1.03)	5.48 (1.18)	64.36 (10.14)	6.81 (2.44)	9.64 (3.89)	34.07 (7.49)	7.02 (2.41)	9.89 (4.14)	35.24 (6.90)	6.64 (2.51)	9.22 (3.66)	33.67 (7.80)	6.70 (2.47)	9.43 (3.96)	33.69 (7.42)					
	Kalman Filter	3.56 (0.92)	5.14 (1.24)	60.34 (9.00)	6.57 (2.34)	9.32 (3.81)	33.20 (6.63)	6.39 (2.65)	9.02 (3.98)	32.21 (8.29)	6.18 (2.48)	8.72 (3.77)	31.52 (8.04)	6.05 (1.98)	8.62 (3.29)	31.14 (5.91)					
	SVR	3.62 (1.03)	5.43 (1.37)	60.77 (7.65)	6.59 (2.21)	9.44 (3.72)	33.29 (6.75)	6.60 (2.29)	9.42 (3.72)	33.28 (7.06)	6.56 (2.47)	9.43 (3.99)	33.08 (7.24)	6.39 (1.97)	9.05 (3.20)	32.76 (5.69)					
	S-UImpute	1.70 (0.51)	3.28 (0.80)	30.30 (4.77)	1.76 (0.47)	3.19 (0.76)	32.73 (2.02)	1.76 (0.54)	3.20 (0.77)	32.68 (3.85)	1.67 (0.54)	3.21 (1.02)	29.66 (3.67)	1.68 (0.45)	3.07 (0.82)	30.62 (2.77)					
Cumhuriyet Bulvarı	ZeroFill	6.09 (2.11)	8.57 (3.32)	31.52 (6.60)	6.66 (2.53)	9.66 (4.23)	33.39 (7.31)	6.89 (2.26)	10.38 (4.20)	34.23 (6.08)	7.06 (2.36)	10.70 (4.36)	35.25 (5.74)	7.32 (2.81)	11.27 (5.30)	35.80 (6.80)					
	MeanFill	6.65 (2.44)	9.40 (3.87)	33.68 (6.86)	6.42 (1.89)	9.19 (3.26)	33.18 (6.30)	7.67 (2.66)	10.44 (4.29)	38.19 (7.01)	7.49 (2.79)	10.42 (4.44)	37.41 (8.71)	7.71 (3.07)	10.79 (4.84)	37.88 (9.27)					
	MedianFill	6.28 (1.95)	8.89 (3.51)	32.39 (5.16)	7.15 (2.62)	9.95 (4.26)	35.81 (7.11)	6.98 (2.37)	9.71 (3.78)	35.63 (7.26)	7.37 (2.50)	10.33 (4.18)	36.98 (7.25)	7.58 (2.80)	10.84 (4.81)	37.33 (8.18)					
	FFill	6.67 (2.32)	9.28 (3.75)	33.89 (6.66)	6.73 (2.08)	9.41 (3.42)	34.40 (6.54)	6.90 (2.03)	9.69 (3.48)	34.81 (6.82)	6.80 (1.95)	9.41 (3.20)	34.65 (4.99)	6.68 (2.49)	9.57 (4.11)	33.33 (6.86)					
	BFill	6.45 (2.19)	9.33 (3.95)	32.88 (6.44)	6.72 (1.60)	9.39 (2.94)	34.43 (5.49)	6.22 (2.34)	8.67 (3.55)	31.97 (7.67)	6.49 (1.99)	9.02 (3.00)	33.48 (6.08)	6.38 (2.20)	8.88 (3.31)	32.61 (6.50)					
	InterpolationFill	6.27 (1.88)	8.73 (3.22)	32.34 (5.49)	6.25 (2.20)	8.75 (3.48)	32.00 (6.67)	6.21 (1.99)	8.98 (3.65)	31.83 (5.94)	6.88 (2.53)	9.63 (4.05)	34.48 (6.59)	6.59 (2.74)	9.26 (4.04)	33.11 (8.93)					
	Kalman Filter	6.30 (2.08)	8.85 (3.31)	32.40 (6.39)	6.76 (2.36)	9.65 (4.28)	34.09 (6.60)	6.35 (1.98)	8.97 (3.25)	32.61 (5.77)	6.62 (2.43)	9.32 (3.69)	33.54 (8.45)	6.89 (2.59)	9.57 (3.98)	34.61 (7.35)					
	SVR	6.60 (2.32)	9.34 (3.87)	33.58 (7.16)	6.53 (1.98)	9.13 (3.31)	33.41 (4.94)	6.58 (2.24)	9.49 (3.73)	33.27 (7.81)	6.27 (1.70)	9.04 (3.37)	32.57 (4.49)	7.01 (2.30)	9.80 (3.71)	35.24 (7.40)					
	S-UImpute	3.54 (0.76)	4.85 (1.12)	19.52 (4.13)	3.37 (0.78)	4.73 (1.29)	18.52 (3.75)	3.33 (0.75)	4.58 (1.19)	18.71 (3.40)	3.10 (0.65)	4.47 (1.23)	16.95 (2.37)	2.98 (0.62)	4.28 (1.04)	16.52 (2.39)					
Odunpazarı	ZeroFill	4.45 (1.43)	5.84 (2.22)	54.09 (25.31)	7.48 (2.68)	11.49 (5.05)	36.82 (6.24)	7.21 (2.46)	10.98 (4.69)	35.71 (6.33)	7.60 (3.08)	11.52 (5.48)	37.31 (7.42)	7.50 (2.76)	11.46 (5.07)	37.02 (6.61)					
	MeanFill	4.67 (1.45)	6.05 (2.05)	56.65 (29.55)	7.80 (2.84)	10.63 (4.35)	38.70 (8.05)	8.01 (2.68)	10.96 (4.21)	39.54 (7.36)	7.94 (2.28)	10.98 (4.02)	39.34 (5.64)	7.89 (2.67)	10.85 (4.43)	39.08 (7.01)					
	MedianFill	4.61 (1.43)	6.11 (2.38)	55.20 (25.06)	7.45 (2.56)	10.63 (4.77)	37.11 (7.04)	7.24 (2.24)	10.09 (3.87)	36.64 (6.55)	7.30 (2.44)	10.39 (4.34)	36.51 (7.29)	7.38 (2.64)	10.63 (4.72)	36.62 (7.55)					
	FFill	5.01 (1.70)	6.38 (2.27)	58.73 (28.51)	6.60 (2.35)	9.42 (3.87)	33.11 (6.22)	6.75 (2.31)	9.58 (3.93)	33.85 (6.02)	6.44 (2.32)	9.22 (3.90)	33.75 (6.86)	6.79 (2.46)	9.77 (4.27)	33.96 (7.02)					
	BFill	4.78 (1.54)	6.23 (2.44)	56.48 (25.39)	6.40 (2.16)	8.96 (3.42)	32.94 (7.08)	6.29 (2.14)	8.88 (3.38)	32.32 (6.73)	6.33 (1.93)	8.81 (3.00)	32.77 (5.62)	6.15 (1.88)	8.56 (2.99)	32.01 (6.46)					
	InterpolationFill	4.34 (1.56)	5.84 (2.56)	51.85 (17.24)	6.77 (2.15)	9.43 (3.49)	34.52 (5.80)	6.42 (2.35)	9.03 (3.44)	32.88 (7.50)	6.62 (1.33)	9.28 (2.70)	34.22 (3.80)	6.65 (2.36)	9.30 (3.57)	33.63 (6.83)					
	Kalman Filter	4.61 (1.84)	5.98 (2.29)	54.80 (27.42)	6.67 (2.41)	9.29 (3.68)	33.87 (7.76)	6.23 (2.07)	8.71 (3.38)	32.10 (6.02)	6.86 (2.17)	9.67 (3.70)	34.68 (6.06)	6.94 (2.30)	9.77 (3.79)	34.77 (6.17)					
	SVR	4.56 (1.23)	5.89 (1.91)	55.95 (26.47)	6.88 (2.31)	9.64 (3.71)	34.81 (7.21)	6.91 (2.58)	9.70 (4.20)	34.51 (7.16)	6.79 (2.33)	9.39 (3.79)	34.44 (6.39)	6.39 (1.92)	9.31 (3.67)	32.63 (5.62)					
	S-UImpute	2.20 (0.70)	3.23 (1.31)	31.05 (10.60)	2.23 (0.91)	3.25 (1.46)	29.86 (14.60)	2.04 (0.80)	3.03 (1.38)	28.96 (11.12)	2.02 (0.88)	3.17 (1.59)	25.63 (8.03)	2.17 (0.98)	3.27 (1.70)	29.92 (13.69)					

(Continued)

Table 7 (continued)

Dataset	Method	Original data					10% Missing					15% Missing					20% Missing					25% Missing						
		MAE	RMSE	msMAPE	MAE	RMSE	msMAPE	MAE	RMSE	msMAPE	MAE	RMSE	msMAPE	MAE	RMSE	msMAPE	MAE	RMSE	msMAPE	MAE	RMSE	msMAPE	MAE	RMSE	msMAPE			
Tepedaşı	ZeroFill	9.87 (2.21)	13.43 (2.70)	20.76 (3.38)	7.26 (2.16)	11.02 (4.38)	36.30 (4.14)	7.39 (2.75)	11.29 (5.13)	36.27 (6.93)	7.60 (2.81)	11.59 (5.20)	37.23 (6.77)	7.34 (2.64)	11.12 (4.84)	36.12 (6.41)	7.60 (2.81)	11.59 (5.20)	37.23 (6.77)	7.34 (2.64)	11.12 (4.84)	36.12 (6.41)	7.60 (2.81)	11.59 (5.20)	37.23 (6.77)	7.34 (2.64)	11.12 (4.84)	36.12 (6.41)
	MeanFill	9.09 (2.30)	12.32 (2.84)	19.12 (3.70)	7.54 (2.81)	10.30 (4.27)	37.73 (8.33)	7.83 (2.79)	10.92 (4.76)	38.64 (8.31)	7.89 (2.54)	10.83 (4.28)	39.02 (6.48)	8.08 (2.31)	10.98 (3.92)	40.01 (5.70)	7.89 (2.54)	10.83 (4.28)	39.02 (6.48)	8.08 (2.31)	10.98 (3.92)	40.01 (5.70)	7.89 (2.54)	10.83 (4.28)	39.02 (6.48)	8.08 (2.31)	10.98 (3.92)	40.01 (5.70)
	MedianFill	9.53 (1.98)	12.83 (2.67)	20.06 (3.00)	7.51 (2.86)	10.64 (4.87)	37.22 (8.08)	7.41 (2.55)	10.61 (4.80)	36.83 (6.55)	7.52 (3.16)	10.61 (4.92)	37.04 (9.42)	7.36 (2.60)	10.53 (4.86)	36.77 (6.62)	7.52 (3.16)	10.61 (4.92)	37.04 (9.42)	7.36 (2.60)	10.53 (4.86)	36.77 (6.62)	7.52 (3.16)	10.61 (4.92)	37.04 (9.42)	7.36 (2.60)	10.53 (4.86)	36.77 (6.62)
	FFill	9.76 (2.37)	13.10 (3.05)	20.46 (3.93)	5.96 (1.83)	8.66 (3.27)	30.67 (5.59)	7.11 (2.30)	9.96 (3.63)	35.47 (6.74)	6.75 (2.17)	9.55 (3.67)	33.88 (5.76)	6.86 (2.52)	9.76 (4.09)	34.20 (6.89)	6.75 (2.17)	9.55 (3.67)	33.88 (5.76)	6.86 (2.52)	9.76 (4.09)	34.20 (6.89)	6.75 (2.17)	9.55 (3.67)	33.88 (5.76)	6.86 (2.52)	9.76 (4.09)	34.20 (6.89)
	BFill	9.02 (1.69)	12.35 (2.51)	18.98 (2.62)	6.37 (2.43)	8.99 (3.78)	32.55 (7.99)	6.42 (2.02)	8.98 (3.26)	32.94 (7.21)	6.76 (2.22)	9.41 (3.66)	34.30 (5.58)	6.46 (2.13)	9.07 (3.49)	33.12 (6.28)	6.76 (2.22)	9.41 (3.66)	34.30 (5.58)	6.46 (2.13)	9.07 (3.49)	33.12 (6.28)	6.76 (2.22)	9.41 (3.66)	34.30 (5.58)	6.46 (2.13)	9.07 (3.49)	33.12 (6.28)
	InterpolationFill	10.01 (2.38)	13.37 (3.02)	21.00 (4.06)	6.35 (1.66)	9.17 (3.25)	32.42 (4.56)	6.49 (2.34)	9.03 (3.61)	33.13 (6.74)	6.57 (1.99)	9.31 (3.44)	33.34 (6.01)	6.46 (2.39)	9.20 (3.89)	32.79 (7.33)	6.57 (1.99)	9.31 (3.44)	33.34 (6.01)	6.46 (2.39)	9.20 (3.89)	32.79 (7.33)	6.57 (1.99)	9.31 (3.44)	33.34 (6.01)	6.46 (2.39)	9.20 (3.89)	32.79 (7.33)
Vişnepark	Kalman Filter	9.49 (1.90)	12.91 (2.67)	19.86 (3.15)	6.65 (2.37)	9.29 (3.69)	33.65 (6.75)	6.49 (2.17)	9.06 (3.53)	33.23 (7.05)	6.21 (2.54)	8.90 (4.06)	31.55 (7.89)	6.34 (2.23)	8.87 (3.54)	32.39 (7.02)	6.21 (2.54)	8.90 (4.06)	31.55 (7.89)	6.34 (2.23)	8.87 (3.54)	32.39 (7.02)	6.21 (2.54)	8.90 (4.06)	31.55 (7.89)	6.34 (2.23)	8.87 (3.54)	32.39 (7.02)
	SVR	9.70 (2.20)	13.13 (2.87)	20.32 (3.79)	6.86 (2.20)	9.67 (3.76)	34.63 (5.61)	6.69 (2.35)	9.53 (3.94)	33.67 (6.69)	6.88 (2.11)	9.61 (3.50)	34.84 (6.23)	6.53 (2.20)	9.18 (3.39)	33.29 (6.14)	6.88 (2.11)	9.61 (3.50)	34.84 (6.23)	6.53 (2.20)	9.18 (3.39)	33.29 (6.14)	6.88 (2.11)	9.61 (3.50)	34.84 (6.23)	6.53 (2.20)	9.18 (3.39)	33.29 (6.14)
	S-Ulmpute	3.79 (0.58)	5.77 (1.05)	8.16 (1.29)	3.99 (0.58)	6.01 (1.11)	8.54 (1.45)	4.01 (0.54)	6.15 (1.41)	8.52 (1.29)	4.00 (0.35)	6.16 (1.20)	8.48 (0.78)	3.91 (0.44)	5.89 (1.05)	8.49 (1.19)	4.00 (0.35)	6.16 (1.20)	8.48 (0.78)	3.91 (0.44)	5.89 (1.05)	8.49 (1.19)	4.00 (0.35)	6.16 (1.20)	8.48 (0.78)	3.91 (0.44)	5.89 (1.05)	8.49 (1.19)
	ZeroFill	12.67 (2.79)	25.58 (9.74)	31.92 (5.67)	12.71 (2.57)	26.09 (10.00)	32.20 (5.01)	13.03 (2.71)	27.23 (9.90)	32.91 (5.56)	12.85 (2.45)	25.93 (7.48)	32.82 (4.98)	13.72 (2.40)	27.39 (7.08)	35.62 (5.84)	12.85 (2.45)	25.93 (7.48)	32.82 (4.98)	13.72 (2.40)	27.39 (7.08)	35.62 (5.84)	12.85 (2.45)	25.93 (7.48)	32.82 (4.98)	13.72 (2.40)	27.39 (7.08)	35.62 (5.84)
	MeanFill	12.16 (2.28)	24.96 (10.11)	30.79 (4.23)	12.72 (2.67)	26.04 (9.66)	32.00 (5.84)	12.88 (2.68)	26.87 (9.85)	32.40 (5.70)	12.69 (2.87)	24.90 (7.81)	32.61 (6.50)	13.29 (3.10)	25.77 (7.92)	34.17 (6.75)	12.69 (2.87)	24.90 (7.81)	32.61 (6.50)	13.29 (3.10)	25.77 (7.92)	34.17 (6.75)	12.69 (2.87)	24.90 (7.81)	32.61 (6.50)	13.29 (3.10)	25.77 (7.92)	34.17 (6.75)
	MedianFill	12.16 (2.99)	25.08 (10.12)	30.54 (6.04)	12.94 (3.15)	26.39 (10.38)	32.63 (6.31)	12.52 (2.86)	25.88 (10.30)	31.60 (6.61)	12.44 (2.36)	24.65 (7.59)	31.94 (5.70)	12.30 (2.47)	25.16 (7.90)	31.34 (5.59)	12.44 (2.36)	24.65 (7.59)	31.94 (5.70)	12.30 (2.47)	25.16 (7.90)	31.34 (5.59)	12.44 (2.36)	24.65 (7.59)	31.94 (5.70)	12.30 (2.47)	25.16 (7.90)	31.34 (5.59)
S-Ulmpute	FFill	12.74 (2.59)	25.66 (9.01)	31.89 (5.96)	13.08 (2.78)	26.21 (9.87)	32.69 (5.76)	12.59 (3.05)	26.31 (10.13)	31.28 (6.59)	12.42 (2.53)	24.50 (6.96)	31.39 (6.05)	12.14 (2.31)	24.68 (7.68)	30.76 (5.54)	12.42 (2.53)	24.50 (6.96)	31.39 (6.05)	12.14 (2.31)	24.68 (7.68)	30.76 (5.54)	12.42 (2.53)	24.50 (6.96)	31.39 (6.05)	12.14 (2.31)	24.68 (7.68)	30.76 (5.54)
	BFill	12.83 (2.94)	25.04 (10.44)	32.53 (6.27)	12.87 (2.74)	25.19 (9.86)	32.69 (7.07)	12.81 (2.49)	25.63 (10.50)	32.46 (5.86)	12.72 (2.64)	23.96 (5.25)	31.93 (7.05)	12.10 (2.89)	23.15 (7.62)	30.87 (6.29)	12.72 (2.64)	23.96 (5.25)	31.93 (7.05)	12.10 (2.89)	23.15 (7.62)	30.87 (6.29)	12.72 (2.64)	23.96 (5.25)	31.93 (7.05)	12.10 (2.89)	23.15 (7.62)	30.87 (6.29)
	InterpolationFill	12.54 (2.30)	25.25 (9.33)	31.41 (5.31)	12.71 (2.91)	25.74 (9.56)	31.90 (6.70)	12.89 (2.94)	25.57 (9.89)	32.41 (6.57)	12.87 (2.75)	24.26 (6.28)	32.48 (6.17)	12.72 (2.55)	23.87 (6.50)	32.52 (6.62)	12.87 (2.75)	24.26 (6.28)	32.48 (6.17)	12.72 (2.55)	23.87 (6.50)	32.52 (6.62)	12.87 (2.75)	24.26 (6.28)	32.48 (6.17)	12.72 (2.55)	23.87 (6.50)	32.52 (6.62)
	Kalman Filter	12.73 (3.00)	25.07 (9.90)	32.10 (5.58)	12.36 (2.59)	24.87 (9.93)	31.30 (6.16)	12.87 (2.95)	26.10 (9.78)	32.06 (6.92)	12.41 (2.32)	24.30 (6.55)	31.40 (5.87)	12.19 (2.24)	24.36 (7.65)	30.82 (5.21)	12.41 (2.32)	24.30 (6.55)	31.40 (5.87)	12.19 (2.24)	24.36 (7.65)	30.82 (5.21)	12.41 (2.32)	24.30 (6.55)	31.40 (5.87)	12.19 (2.24)	24.36 (7.65)	30.82 (5.21)
	SVR	12.99 (3.09)	25.63 (10.31)	32.60 (6.70)	12.72 (2.90)	25.37 (10.01)	32.01 (6.11)	12.58 (2.87)	25.61 (9.65)	31.48 (6.26)	12.57 (2.72)	24.34 (6.93)	31.82 (6.36)	11.97 (2.33)	24.37 (7.46)	30.40 (5.65)	12.57 (2.72)	24.34 (6.93)	31.82 (6.36)	11.97 (2.33)	24.37 (7.46)	30.40 (5.65)	12.57 (2.72)	24.34 (6.93)	31.82 (6.36)	11.97 (2.33)	24.37 (7.46)	30.40 (5.65)
	S-Ulmpute	8.20 (1.21)	18.95 (7.25)	21.46 (2.43)	8.47 (1.72)	19.45 (9.03)	22.14 (3.88)	8.65 (2.13)	21.72 (13.03)	21.89 (3.22)	7.88 (1.31)	17.42 (6.36)	20.33 (2.53)	7.77 (1.40)	19.07 (9.83)	19.95 (2.04)	7.88 (1.31)	17.42 (6.36)	20.33 (2.53)	7.77 (1.40)	19.07 (9.83)	19.95 (2.04)	7.88 (1.31)	17.42 (6.36)	20.33 (2.53)	7.77 (1.40)	19.07 (9.83)	19.95 (2.04)

Note:
The bold values indicate the best performance for each metric, corresponding to the lowest MAE, RMSE, and msMAPE values.

Table 8 Average ranks of imputation methods based on the Friedman test (cross-validation results).

Method	Average rank
S-UImpute	1.12
Kalman	3.88
BFill	4.00
Interpolation	4.64
FFill	5.12
SVR	5.16
ZeroFill	6.68
MedianFill	6.80
MeanFill	7.60

values across datasets, which proves that the model is more stable and reliable even when the missing data rate increases. Taken together, these results demonstrate that S-UImpute provides a more stable and accurate basis for LSTM forecasting across different stations and missing data levels.

Statistical analysis

To statistically evaluate the effect of missing data imputation methods on forecasting performance, we first performed the Friedman test.

The Friedman test is suitable for evaluating the performance of K different models on N different datasets (Rainio, Teuho & Klén, 2024). In the test statistic in Eq. (16), N is the number of data sets, K is the number of models, and $\bar{R}_k$ is the average rank of each model over all datasets based on the evaluation metric.

$$\chi_F^2 = \frac{12N}{K(K+1)} \left[\sum_{k=1}^K \bar{R}_k^2 - \frac{K(K+1)^2}{4} \right]. \quad (16)$$

The null hypothesis and the alternative hypothesis are defined as follows:

H_0 : There is no significant difference among the imputation methods.

H_1 : There is a significant difference among the imputation methods.

Here, the performances of the forecasting models created using nine different missing data imputation methods were tested on 25 different datasets. The Friedman test was applied to the msMAPE values obtained from the five-fold cross-validation results of the LSTM model, which was the most successful forecasting model. The average success rank, based on the msMAPE rank of the methods, is given in the Table 8 for the Friedman test. Lower ranks indicate better performance. S-UImpute achieved the best overall performance in terms of average rank. As a result of the Friedman test performed at a 95% confidence level, the p -value is smaller than 0.001, and the test statistic χ^2 is 101.003. Since the p -value is less than 0.05, the null hypothesis is rejected. This indicates that missing data imputation methods are statistically different in terms of forecasting success.

Table 9 Pairwise comparison results between S-UImpute and other imputation methods using the Wilcoxon signed-rank test with Holm-Bonferroni correction (cross-validation results).

Compared methods	Corrected p -value	H_0
S-UImpute vs. ZeroFill	4.77×10^{-7}	Reject
S-UImpute vs. MeanFill	4.77×10^{-7}	Reject
S-UImpute vs. MedianFill	4.77×10^{-7}	Reject
S-UImpute vs. FFill	4.77×10^{-7}	Reject
S-UImpute vs. BFill	4.77×10^{-7}	Reject
S-UImpute vs. Interpolation	4.77×10^{-7}	Reject
S-UImpute vs. Kalman	4.77×10^{-7}	Reject
S-UImpute vs. SVR	4.77×10^{-7}	Reject

If the Friedman test indicates a statistically significant difference, *post-hoc* analysis is required to determine the specific pairs of methods that exhibit significant differences (Demsar, 2006). Following the Friedman test, *post-hoc* pairwise comparisons are performed using the Wilcoxon signed-rank test with Holm-Bonferroni correction, as recommended by Benavoli, Corani & Mangili (2016). The Wilcoxon signed-rank test evaluates whether two related methods systematically differ in performance by ranking the paired differences across datasets, while the Holm-Bonferroni correction adjusts the significance thresholds to control the family-wise error rate in multiple pairwise comparisons (Aula & Rashid, 2025). After demonstrating significant differences among the imputation methods with the Friedman test, the proposed S-UImpute was compared with the other methods using the Wilcoxon signed-rank test with Holm-Bonferroni correction. The null hypothesis and alternative hypothesis:

H_0 : There is no difference between the compared imputation methods.

H_a : The compared imputation methods are significantly different from each other.

Table 9 shows the results of the Wilcoxon signed-rank test with Holm-Bonferroni correction test at the 95% confidence level. According to the results, H_0 is rejected for all pairwise comparisons. This implies that S-UImpute is significantly different from all of the performed imputation methods.

DISCUSSION

The results demonstrate that the choice of imputation method significantly affects air quality forecasting. Our proposed S-UImpute method, especially with LSTM, provided lower msMAPE values and more stable forecasts than other approaches. Simple methods like ZeroFill and MeanFill get much worse as the missing data rate increases. On the other hand, methods like interpolation and Kalman filtering perform better than simple methods, but they are still not robust enough. Our findings are in line with (Junninen et al., 2004) emphasizing that the choice of imputation method plays a critical role in air-quality forecasting accuracy.

A five-fold cross-validation with an expanding window was conducted to examine the model robustness. The cross-validation results show that S-UImpute combined with

LSTM offers high accuracy and low variability, even with different amounts of missing data. According to the Friedman test, there are significant differences among the imputation methods. S-UImpute achieved the best average rank, and Wilcoxon signed-rank as a *post-hoc* test proved that it is significantly better than the other methods. Consistently, our findings support (Aleryani et al., 2023), who demonstrated that ensemble-based imputation methods significantly improve the robustness and predictive performance of time series models compared to single imputation techniques.

This study contributes to the literature by introducing S-UImpute, a novel missing data imputation method designed to improve forecast performance under varying missing data rates. S-UImpute offers an important advantage by combining the strengths of different imputation strategies with a stacking ensemble framework. Because individual methods tend to behave inconsistently across stations and under varying missing data ratios, S-UImpute was designed to produce more stable and reliable estimates, which is reflected in its superior performance throughout the experiments. Several limitations should be acknowledged. The current analysis is limited to univariate time series. Incorporating meteorological, traffic, or spatial variables in future extensions may provide a more comprehensive forecasting structure. Additionally, exploring different missing data mechanisms and rates could help further assess the robustness and generalizability of the S-UImpute method. Finally, the analysis is based on five stations in one city and three commonly used forecasting models (ARIMA, SVR, and LSTM), the findings should be interpreted with the understanding that results may vary across regions or forecasting models.

CONCLUSIONS

Taking precautions against air pollutants is an effective way to reduce health, environmental, and economic impacts in Eskişehir, Türkiye, where industrial activities and other factors lead to high pollutant exposure. Therefore, developing potential air pollutant forecasting models will be a valuable tool for decision-makers. Using data from five air quality stations in Eskişehir, Türkiye, we built forecasting models in this study utilizing statistical, machine learning, and deep learning models. We propose a novel stacking ensemble imputation method, S-UImpute, to address the missing data problem that arises in real-world data and is also present in acquired datasets.

In this context, we compared the effect of S-UImpute on forecasting performance with existing imputation methods for univariate time series. When the test performance of each imputation method and forecasting model combination was compared, the S-UImpute + LSTM model mostly outperformed the other models. Although there are models that outperform the S-UImpute + LSTM model in a few datasets, the generalized and successful forecasting performance of the proposed approach will allow it to be preferred over other models. In addition to the original data, the S-UImpute + LSTM model outperformed the other models in datasets where the missing data rate was increased. The statistical tests performed also support these findings. The results of the statistical tests indicate that the S-UImpute method differs from existing univariate imputation methods. According to the average ranking obtained, S-UImpute achieved the best overall performance.

For future work, we plan to extend the study by moving from univariate forecasting to a multivariate and spatiotemporal structure. Including multiple pollutants together with external factors such as weather and traffic variables may improve forecast accuracy and provide a better understanding of how air pollution behaves across different parts of the city. In addition, S-UImpute can be tested within more recent transformer-based models, which have shown strong performance over longer time periods and in spatiotemporal forecasting.

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ADDITIONAL INFORMATION AND DECLARATIONS

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Competing Interests

The authors declare that they have no competing interests.

Author Contributions

- Zeynep Idil Erzurum Cicek conceived and designed the experiments, performed the experiments, analyzed the data, performed the computation work, prepared figures and/or tables, authored or reviewed drafts of the article, and approved the final draft.
- Zeliha Ergul Aydin conceived and designed the experiments, performed the experiments, analyzed the data, performed the computation work, prepared figures and/or tables, and approved the final draft.

Data Availability

The following information was supplied regarding data availability:

The data is available at GitHub:

- <https://github.com/Zierzurum/airpollution>.

- Erzurum Cicek, Z. I., & Ergul Aydin, Z. (2025). Air Pollution Dataset-Eskişehir (2024–2025) (v0.2). Zenodo. <https://doi.org/10.5281/zenodo.17276821>.

The code is available at GitHub:

- <https://github.com/Zierzurum/S-UImpute>.

- Erzurum Cicek, Z. I., & Ergul Aydin, Z. (2025). S-UImpute (v.0.2). Zenodo. <https://doi.org/10.5281/zenodo.17276793>.

REFERENCES

- Aleryani A, Bostrom A, Wang W, Iglesia B. 2023.** Multiple imputation ensembles for time series (MIE-TS). *ACM Transactions on Knowledge Discovery from Data* **17(3)**:1–28 DOI [10.1145/3551643](https://doi.org/10.1145/3551643).
- Aleryani A, Wang W, De La Iglesia B. 2020.** Multiple imputation ensembles (MIE) for dealing with missing data. *SN Computer Science* **1(3)**:134 DOI [10.1007/s42979-020-00131-0](https://doi.org/10.1007/s42979-020-00131-0).
- Alkabbani H, Ramadan A, Zhu Q, Elkamel A. 2022.** An improved air quality index machine learning-based forecasting with multivariate data imputation approach. *Atmosphere* **13(7)**:1144 DOI [10.3390/atmos13071144](https://doi.org/10.3390/atmos13071144).
- Alsaber AR, Pan J, Al-Hurban A. 2021.** Handling complex missing data using random forest approach for an air quality monitoring dataset: a case study of Kuwait environmental data (2012 to 2018). *International Journal of Environmental Research and Public Health* **18(3)**:1333 DOI [10.3390/ijerph18031333](https://doi.org/10.3390/ijerph18031333).
- Aula SA, Rashid TA. 2025.** FOX-TSA: navigating complex search spaces and superior performance in benchmark and real-world optimization problems. *Ain Shams Engineering Journal* **16(1)**:103185 DOI [10.1016/j.asej.2024.103185](https://doi.org/10.1016/j.asej.2024.103185).
- Bakar MAA, Ariff NM, Man PH, Nadzir MSM. 2024.** Forecasting the air quality time series data using univariate and multivariate convolutional neural network models. In: *AIP Conference Proceedings*. Vol. 2895. Melville: AIP Publishing.
- Balcılar İ. 2024.** Eskişehir’de hava kirliliği: PM10, PM2.5 ve SO2 konsantrasyonlarının mekânsal-zamansal değişimi ve kaynaklarının değerlendirilmesi. *Niğde Ömer Halisdemir Üniversitesi Mühendislik Bilimleri Dergisi* **13(4)**:1115–1126 DOI [10.28948/ngumuh.1459990](https://doi.org/10.28948/ngumuh.1459990).
- Benavoli A, Corani G, Mangili F. 2016.** Should we really use post-hoc tests based on mean-ranks? *Journal of Machine Learning Research* **17(5)**:1–10.
- Bentéjac C, Csörgő A, Martínez-Muñoz G. 2021.** A comparative analysis of gradient boosting algorithms. *Artificial Intelligence Review* **54(3)**:1937–1967 DOI [10.1007/s10462-020-09896-5](https://doi.org/10.1007/s10462-020-09896-5).
- Bergmeir C, Benítez JM. 2012.** On the use of cross-validation for time series predictor evaluation. *Information Sciences* **191(3)**:192–213 DOI [10.1016/j.ins.2011.12.028](https://doi.org/10.1016/j.ins.2011.12.028).
- Box G, Jenkins G. 1970.** Time series analysis: forecasting and control. In: *Holden-Day Series in Time Series Analysis and Digital Processing*. San Francisco: Holden-Day.
- Cicek ZIE. 2024.** A stacking ensemble model for forecasting solar photovoltaic power generation. In: *2024 Innovations in Intelligent Systems and Applications Conference (ASYU)*. Piscataway: IEEE, 1–5.
- Çiçek ZIE, Aydın ZE. 2025.** Air pollution dataset-eskişehir (2024–2025). Available at <https://github.com/Zierzurum/airpollution>.
- Cicek ZIE, Aydın ZE. 2025.** Air quality forecasting in presence of missing data. In: Bennour A, Bouridane A, Almaadeed S, Bouaziz B, Edirisinghe E, eds. *Intelligent Systems and Pattern Recognition*. Cham: Springer Nature Switzerland, 250–261.
- Cortes C, Vapnik V. 1995.** Support-vector networks. *Machine Learning* **20(3)**:273–297 DOI [10.1023/a:1022627411411](https://doi.org/10.1023/a:1022627411411).

- Demsar J. 2006.** Statistical comparisons of classifiers over multiple data sets. *Journal of Machine Learning Research* 7(1):1–30.
- Diniz APM, Ciarelli PM, Salles EOT, Coco KF. 2024.** Use of deep neural networks for clogging detection in the submerged entry nozzle of the continuous casting. *Expert Systems with Applications* 238(6):121963 DOI 10.1016/j.eswa.2023.121963.
- Drucker H, Burges CJC, Kaufman L, Smola A, Vapnik V. 1996.** Support vector regression machines. In: Mozer M, Jordan M, Petsche T, eds. *Advances in Neural Information Processing Systems*. Vol. 9. USA: MIT Press.
- Du W, Côté D, Liu Y. 2023.** SAITS: self-attention-based imputation for time series. *Expert Systems with Applications* 219(10):119619 DOI 10.1016/j.eswa.2023.119619.
- Eren B, Aksangür İ, Erden C. 2023.** Predicting next hour fine particulate matter (PM_{2.5}) in the Istanbul metropolitan city using deep learning algorithms with time windowing strategy. *Urban Climate* 48:101418 DOI 10.1016/j.uclim.2023.101418.
- Essamlali I, Nhaila H, El Khaili M. 2024.** Supervised machine learning approaches for predicting key pollutants and for the sustainable enhancement of urban air quality: a systematic review. *Sustainability* 16(3):976 DOI 10.3390/su16030976.
- Hadeed SJ, O'Rourke MK, Burgess JL, Harris RB, Canales RA. 2020.** Imputation methods for addressing missing data in short-term monitoring of air pollutants. *Science of the Total Environment* 730:139140 DOI 10.1016/j.scitotenv.2020.139140.
- Hochreiter S, Schmidhuber J. 1997.** Long short-term memory. *Neural Computation* 9(8):1735–1780 DOI 10.1162/neco.1997.9.8.1735.
- Hua V, Nguyen T, Dao MS, Nguyen HD, Nguyen BT. 2024.** The impact of data imputation on air quality prediction problem. *PLOS ONE* 19(9):e0306303 DOI 10.1371/journal.pone.0306303.
- Hyndman RJ, Athanasopoulos G. 2021.** *Forecasting: principles and practice*. Third Edition. Melbourne, Australia: OTexts.
- Jäger S, Allhorn A, Bießmann F. 2021.** A benchmark for data imputation methods. *Frontiers in Big Data* 4:693674 DOI 10.3389/fdata.2021.693674.
- Junninen H, Niska H, Tuppurainen K, Ruuskanen J, Kolehmainen M. 2004.** Methods for imputation of missing values in air quality data sets. *Atmospheric Environment* 38(18):2895–2907 DOI 10.1016/j.atmosenv.2004.02.026.
- Kangas T, Gadeyne S, Lefebvre W, Vanpoucke C, Rodriguez-Loureiro L. 2023.** Are air quality perception and PM_{2.5} exposure differently associated with cardiovascular and respiratory disease mortality in Brussels? Findings from a census-based study. *Environmental Research* 219(1):115180 DOI 10.1016/j.envres.2022.115180.
- Kingphai K, Moshfeghi Y. 2023.** On time series cross-validation for deep learning classification model of mental workload levels based on EEG signals. In: Nicosia G, Ojha V, La Malfa E, La Malfa G, Pardalos P, Di Fatta G, Giuffrida G, Umeton R, eds. *Machine Learning, Optimization, and Data Science*. Cham: Springer Nature Switzerland, 402–416.
- Lee J-Y, Han S-H, Kang J-G, Lee C-Y, Lee J-B, Kim H-S, Yun H-Y, Choi D-R. 2025.** Comparison of models for missing data imputation in PM-2.5 measurement data. *Atmosphere* 16(4):438 DOI 10.3390/atmos16040438.
- Lewis CD. 1982.** *Industrial and business forecasting methods: a practical guide to exponential smoothing and curve fitting*. London: Butterworths.
- Liang Y, Xia Y, Ke S, Wang Y, Wen Q, Zhang J, Zheng Y, Zimmermann R. 2023.** Airformer: predicting nationwide air quality in China with transformers. *Proceedings of the AAAI Conference on Artificial Intelligence* 37:14329–14337 DOI 10.1609/aaai.v37i12.26676.

- Mani G, Viswanadhapalli JK, Stonier AA. 2022. Prediction and forecasting of air quality index in Chennai using regression and ARIMA time series models. *Journal of Engineering Research* 10(2):179–194 DOI 10.36909/jer.10253.
- Mao W, Wang W, Jiao L, Zhao S, Liu A. 2021. Modeling air quality prediction using a deep learning approach: method optimization and evaluation. *Sustainable Cities and Society* 65:102567 DOI 10.1016/j.scs.2020.102567.
- Méndez M, Merayo MG, Núñez M. 2023. Machine learning algorithms to forecast air quality: a survey. *Artificial Intelligence Review* 56:10031–10066 DOI 10.1007/s10462-023-10424-4.
- Moritz S, Sardá A, Bartz-Beielstein T, Zaefferer M, Stork J. 2015. Comparison of different methods for univariate time series imputation in R. ArXiv DOI 10.48550/arXiv.1510.03924.
- Narkhede G, Poonawala M, Sonawane A, Hiwale A, Singh AR. 2025. Air pollution prediction with advanced preprocessing and deep ensemble learning. *Atmospheric Pollution Research* 16(10):102610 DOI 10.1016/j.apr.2025.102610.
- Niako N, Melgarejo JD, Maestre GE, Vatcheva KP. 2024. Effects of missing data imputation methods on univariate blood pressure time series data analysis and forecasting with ARIMA and LSTM. *BMC Medical Research Methodology* 24(1):320 DOI 10.1186/s12874-024-02448-3.
- Park S, Jung S, Lee J, Hur J. 2023. A short-term forecasting of wind power outputs based on gradient boosting regression tree algorithms. *Energies* 16(3):1132 DOI 10.3390/en16031132.
- Rainio O, Teuho J, Klén R. 2024. Evaluation metrics and statistical tests for machine learning. *Scientific Reports* 14(1):6086 DOI 10.1038/s41598-024-56706-x.
- Ravindiran G, Hayder G, Kanagarathinam K, Alagumalai A, Sonne C. 2023. Air quality prediction by machine learning models: a predictive study on the Indian coastal city of Visakhapatnam. *Chemosphere* 338:139518 DOI 10.1016/j.chemosphere.2023.139518.
- Rizkallah LW. 2025. Enhancing the performance of gradient boosting trees on regression problems. *Journal of Big Data* 12(1):35 DOI 10.1186/s40537-025-01071-3.
- Saihood Q, Sonuç E. 2023. A practical framework for early detection of diabetes using ensemble machine learning models. *Turkish Journal of Electrical Engineering and Computer Sciences* 31(4):722–738 DOI 10.55730/1300-0632.4013.
- Sethi JK, Mittal M. 2020. Analysis of air quality using univariate and multivariate time series models. In: *2020 10th International Conference on Cloud Computing, Data Science & Engineering (Confluence)*, 823–827.
- Shcherbakov MV, Brebels A, Shcherbakova NL, Tyukov AP, Janovsky TA, Kamaev VA. 2013. A survey of forecast error measures. *World Applied Sciences Journal* 24(24):171–176.
- Tagil Ş, Mentese S. 2012. Zonguldak'ta hava kirliliği PM10 & SO2 ile ilişkili olarak seçilmiş solunum yolu hastalıklarının zamansal ve mekansal değişimi. *Balıkesir Üniversitesi Sosyal Bilimler Enstitüsü Dergisi* 15(27):3–18.
- Tawakuli A, Havers B, Gulisano V, Kaiser D, Engel T. 2025. Survey: time-series data preprocessing: a survey and an empirical analysis. *Journal of Engineering Research* 13(2):674–711 DOI 10.1016/j.jer.2024.02.018.
- Turkish Statistical Institute. 2023. Turkish Statistical Institute data portal. Available at <https://veriportal.tuik.gov.tr/en/>. (accessed 15 March 2025).
- Umar N, Gray A. 2023. Comparing single and multiple imputation approaches for missing values in univariate and multivariate water level data. *Water* 15(8):1519 DOI 10.3390/w15081519.
- Wang X, Zhang S, Chen Y, He L, Ren Y, Zhang Z, Li J, Zhang S. 2024. Air quality forecasting using a spatiotemporal hybrid deep learning model based on VMD–GAT–BiLSTM. *Scientific Reports* 14(1):17841 DOI 10.1038/s41598-024-68874-x.

- World Health Organization.** 2025. Air pollution. Available at <https://www.who.int/health-topics/air-pollution> (accessed 25 June 2025).
- Yu C, Tan J, Cheng Y, Mi X.** 2024. Data analysis and preprocessing techniques for air quality prediction: a survey. *Stochastic Environmental Research and Risk Assessment* **38**(6):2095–2117 DOI [10.1007/s00477-024-02693-4](https://doi.org/10.1007/s00477-024-02693-4).
- Yu C, Wang F, Shao Z, Sun T, Wu L, Xu Y.** 2023. DSformer: a double sampling transformer for multivariate time series long-term prediction. In: *Proceedings of the 32nd ACM International Conference on Information and Knowledge Management*, 3062–3072.
- Yu C, Wang F, Wang Y, Shao Z, Sun T, Yao D, Xu Y.** 2025. MGSFformer: a multi-granularity spatiotemporal fusion transformer for air quality prediction. *Information Fusion* **113**:102607 DOI [10.1016/j.inffus.2024.102607](https://doi.org/10.1016/j.inffus.2024.102607).
- Zhang Z, Zhang S, Chen C, Yuan J.** 2024. A systematic survey of air quality prediction based on deep learning. *Alexandria Engineering Journal* **93**:128–141 DOI [10.1016/j.aej.2024.03.031](https://doi.org/10.1016/j.aej.2024.03.031).
- Zhang Z, Zhao Y, Canes A, Steinberg D, Lyashevskaya O.** 2019. Predictive analytics with gradient boosting in clinical medicine. *Annals of Translational Medicine* **7**(7):152 DOI [10.21037/atm.2019.03.29](https://doi.org/10.21037/atm.2019.03.29).
- Zhou X, Liu X, Lan G, Wu J.** 2021. Federated conditional generative adversarial nets imputation method for air quality missing data. *Knowledge-Based Systems* **228**(1):107261 DOI [10.1016/j.knosys.2021.107261](https://doi.org/10.1016/j.knosys.2021.107261).
- Zierzurum Y.** 2025. S-UImpute: stacking-based imputation for air quality forecasting. Available at <https://github.com/Zierzurum/S-UImpute>.